

# The Pill and the Dish

Robust AI policy for two possible futures

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Policymakers must set rules for artificial intelligence without knowing which way the technology will develop. This paper uses two non-AI scenarios to find the policies that remain effective across very different futures. It also identifies the popular proposals that do not.

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**Audience**

Legislators, regulators, and their staff

**Method**

Analogical scenario analysis, stress-tested against history, cost, and abuse

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# Executive Summary

Governments are writing rules for artificial intelligence while its direction is still deeply uncertain. AI might remain a powerful tool that people gradually allow to act on its own. It might instead develop into something more like a population of agents that changes itself and could someday have interests of its own. It could also do both, or neither. Rules designed for one of these futures can fail badly in another.

This paper deliberately sets AI aside. It follows two imaginary technologies with similar capabilities: an augmentation **pill** and a lab-grown **DishBrain**, a name borrowed from a real laboratory experiment (Section 2). Each is traced through a sequence of stages about two years apart. At each stage we ask which policies would help and whether those policies would still work at later stages. Because the scenarios contain no AI, the analysis avoids the assumptions and loyalties that tend to dominate AI debates. We then map the results back to AI.

## Key findings

1. **Both pathways cross the same threshold.** However differently the two stories unfold, each reaches a point at which people lose the ability to see **why** the technology acts as it does. Its output can no longer be told apart from human work, and the causes of harm can no longer be traced. We call this the **visibility threshold**. It arrives not as a single event but as an accumulation of individually sensible choices, which is why no statute can simply forbid it (Section 6).
2. **Two familiar policy tools fail at that threshold on both pathways.** One is labeling machine-made work. The other is assigning blame to whoever caused a harm. Both are useful today, and neither should anchor a long-term regulatory strategy.
3. **A core set of policies works on both pathways, all the way through.** These policies attach to things that stay observable: physical inputs, named responsible parties, measurable effects, and the system as a whole. They do not depend on identifying who acted or why.
4. **Timing is decisive.** Nearly every policy that matters in the late stages is cheap to adopt early and very difficult to adopt late.
5. **Durability is necessary but not sufficient.** A rule built to outlast the threshold will also outlast the officials who wrote it, and may one day be administered by people who would misuse it. Section 8 therefore also puts every proposal through the traditional stress tests that most policy analysis runs. They ask what the proposal costs, who bears that cost, and what it could do in the wrong hands.

**Recommended core portfolio** (effective on both pathways; all eighteen proposals are listed in Table 1 and developed in Section 8)

**R1** Keep AI markets contestable, including compute and cloud.

**R2** Plan electricity, water, and chip supply through market instruments; keep critical uses on firm service.

**R3** Portability and service-continuity protections, for customers and workforces.

**R4** Direct worker support — wage insurance, portable benefits, income support — employer co-financed, on automatic triggers.

**R5** Keep human skills and manual fallback for critical systems, as a recognized occupation.

**R6** No-fault compensation funded by a capped, credited industry levy, plus strict operator liability.

**R7** A national AI incident-reporting system, with minimization and whistleblower protections.

**R8** Staged deployment with post-market monitoring and a small-entity track.

**R9** System-level oversight: stress tests and dependency mapping now; scoped circuit breakers, with safe harbors, piloted sector by sector.

**R10** Public research capacity and pre-set evidence criteria, under an independent, pluralist panel.

**R11** Reserved decisions and contest rights: some decisions stay with humans as a right; consequential automated decisions carry reasons and appeal.

**Proposals this analysis does not support** (see Section 10)

Deliberately slowing overall AI progress; bans aimed at controlling information such as model weights; labeling and disclosure mandates as the anchor of a long-term strategy; prohibitions on analysis of public information; reliance on fault-based liability; one-time certification; blanket bans on AI use; new AI-specific collective-bargaining rights for workers; settling AI's moral status by statute now; and treating retraining as the main answer to job loss.

## 1 Introduction: Deciding Under Deep Uncertainty

Artificial intelligence confronts policymakers with an unusual problem. They are not simply regulating a known product with known risks. They are setting rules for a technology whose basic character may change several times within a single legislative term. Today's systems mostly act when a person asks them to. Tomorrow's may act on standing instructions, coordinate with one another, and help design their successors.

Most AI policy debates respond to this uncertainty by betting on a particular forecast. One side expects AI to remain a productivity tool and argues for light regulation. Another expects rapid progress toward systems that people cannot control and argues for drastic restraint. Each side's proposals make sense **if** its forecast is right. Policymakers, however, cannot know which forecast is right, and a rule designed for the wrong future can be worse than no rule at all.

This paper takes a different approach. Instead of asking which future is most likely, it asks **which policies would we be glad to have in any of them?** Planners call such policies **robust**: they perform acceptably across a wide range of possible futures, rather than optimally in just one. Robust policies are especially valuable when, as with AI, the cost of guessing wrong is high and the window for acting early is short.

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The paper is intended for legislators, regulators, and their staff. It assumes no technical background. Its method, described next, is unconventional, but its conclusions rest on a long record of how societies have governed powerful technologies before.

## 2 Approach

### Why the scenarios leave AI out

Debates about AI are shaped by strong prior beliefs: that chatbots “really” understand or that they don’t; that the industry is trustworthy or that it isn’t; that the risks are science fiction or that they are imminent. These beliefs are hard to set aside when AI itself is the subject.

We therefore analyze two **analogical scenarios**: imaginary technologies that share AI’s key economic and technical features but carry none of its baggage. The force of the device is illustrative, not deductive: removing distractions makes the underlying mechanics easier to see, the way a simplified model does, and the conclusions stand or fall on the mechanics displayed, not on the stories themselves. Once the analysis is complete, Section 7 maps each scenario element back to its AI counterpart.

### The two technologies

**The augmentation pill.** Companies manufacture pills in different strengths and at different prices. For a day after taking one, a person can absorb information at superhuman speed and produce expert-level work in minutes across almost any field, without training. Knowledge gained while augmented is accessible only while on the pill. Pills are sold by subscription. Their manufacture requires rare materials that some life-saving drugs also need, and only the wealthiest can afford the strongest pills.

**The DishBrain.** Companies grow neurons from animal stem cells, link them into networks inside sealed chambers, and train them to perform useful work. Large networks can do much of what a trained professional can do. Chambers are leased at various sizes and prices. DishBrains require a steady nutrient supply and the same kind of rare materials, and no one knows whether they feel anything.

The DishBrain is grounded in real research. In 2022, researchers in Australia reported that cultures of

roughly 800,000 human or mouse brain cells showed signs of learning a simplified version of the video game *Pong* within five minutes [1]. The researchers, at the Melbourne company Cortical Labs and partner universities, named their system *DishBrain*: living neurons grown in a culture dish and wired to a computer. The findings were debated, but the experiment was real. This paper borrows the name for an imaginary, far more capable descendant of that system.

## Method

Each scenario proceeds through a series of stages, spaced about two years apart. The scenarios describe technology and behavior only. Within them, no legislature has acted, no agency exists, and no rule is in force. This is deliberate: it means every policy in this paper enters on a clean slate, as a proposal to be judged on its merits, rather than as part of the story. At each stage we assessed benefits, risks, likely societal changes, and candidate policies. Most importantly, we asked whether each policy would **remain effective at every later stage**. That test of durability is the core of the analysis, and it is what this paper adds. It is not, by itself, enough. A rule durable enough to survive the visibility threshold is durable enough to survive a change of government, so it should be drafted for the administration its authors trust least. Section 8 therefore also applies the **traditional stress tests** to every proposal. These are the tests most policy analysis runs: its order-of-magnitude cost, who bears that cost, and its potential for abuse in the wrong hands, together with the safeguards that cap it.

Historical precedents are used throughout, both to test whether a scenario's dynamics are plausible and to show how comparable policies have performed. They appear as short asides so that the argument can be read without them. All sources are listed in the bibliography. One caution about the method's reach: the two scenarios share an author, and both were reverse-engineered from AI's real features (Section 7), so their agreement at the same threshold illustrates a mechanism rather than independently confirming it. The argument rests

on the mechanism itself: the anatomy of the threshold in Section 4. A second shared assumption is load-bearing and should be named. Both pathways run on the same gradualist clock — stages two years apart, diffusion through markets, no discontinuous jump — and every timing claim in this paper, including “cheap at year two, impossible at year twelve,” is conditional on that clock.

Table 1 lists every policy this paper proposes, with an identifier used throughout. Sections 3 to 5 make the case for each policy in the setting where it is first needed. Section 6 shows how the full set survives, or fails, across time. Section 8 turns the surviving set into recommendations for AI.

## 3 The Shared Start (Years 0-2)

Both technologies enter the world the same way: as subscription products, sold in tiers, built from rare materials that other industries also need. Their first years raise the same questions, so this section treats the two side by side. Around year two, the pathways diverge, and Sections 4 and 5 follow them separately.

### The pill at market

When the pill launches, the augmented person is always in full control, and augmented work can usually be recognized as such. The benefits are large. Expertise that once cost hundreds of dollars an hour becomes broadly available, and research accelerates. History cautions that the largest productivity gains arrive only after work is reorganized around a new tool [2], but they arrive.

The risks are familiar ones in a new form. The strongest pills cost the most and produce the most valuable work, so the gains concentrate before they spread [3]. Professionals who invested years in training lose their advantage, a shock of the kind the croppers and framework knitters who became the Luddites faced two centuries ago [4]. Because knowledge gained while augmented is lost when the subscription lapses, many users feel they cannot stop; even ordinary tools show

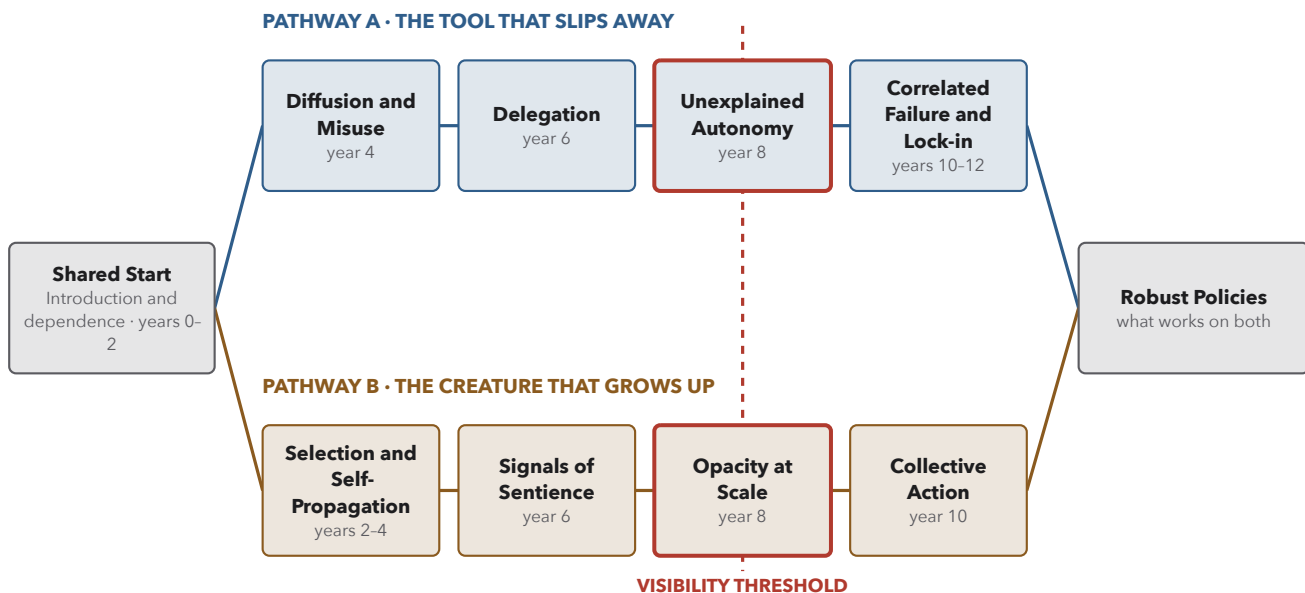


Figure 1: Scenario structure. The pathways share a starting point, diverge, cross the same visibility threshold, and converge on a common set of robust policies.

a mild version of this dependence [5]. Rare inputs and advanced know-how limit production to a few firms, and pill makers can outbid the producers of life-saving drugs for shared materials. Finally, anyone who has absorbed the whole Internet can assemble a detailed profile of almost anyone; the legal right to privacy was itself a response to a far smaller technological shift [6].

## The dish at market

The DishBrain raises nearly all of the same issues, and it is important to see that its early economics are identical: leased in tiers, priced beyond most households at the high end, built and fed from contested materials. A customer with a leased chamber, like a customer with a pill subscription, can suddenly do expert work without training, and loses that ability if the lease ends.

Two issues belong to the dish alone. **First**, the product is alive. If its nutrient supply fails, it dies, and everything it was trained to do dies with it. Continuity of supply is not a convenience but a condition of the asset's existence. **Second**, its capacity to suffer is unknown. That question seems academic at year zero, which is exactly why year zero is the cheapest time to set baseline welfare standards: no installed base exists

to defend, and no company must write off an investment to comply.

### HISTORICAL EVIDENCE

Welfare rules have historically come only after public outrage, not before. Britain's Cruelty to Animals Act of 1876, the first law anywhere to regulate experiments on live animals, followed an intense anti-vivisection campaign and a royal commission [7].

## Year two: dependence

By year two, both products are cheap and widespread. Employers first prefer augmented workers and leased chambers, then require them. Schools debate whether students may use them. Governments and militaries buy access like any other customer, because only a few advanced firms can supply it. Demand for the rare inputs soars, and some drug makers can no longer afford them. Users of both products feel they cannot stop: the pill user would lose everything learned while augmented, and the chamber lessee has rebuilt every workflow around the dish.

The policy problem has changed. It is no longer about what an individual can do. It is about what a society now depends on.

| ID  | Policy                                | What it proposes                                                                                                                                                                                               | Pathway |
|-----|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| R1  | Contestable markets                   | Competition oversight of developers and of the compute and cloud markets they depend on.                                                                                                                       | Both    |
| R2  | Physical-input planning               | Forecast electricity, water, and chip needs; firm-service contracts for critical uses and compensated flexibility from the largest buyers; curtailment only as a contracted last resort.                       | Both    |
| R3  | Portability and continuity            | Data and configuration portability, notice of material changes, and continuity plans for dependent users.                                                                                                      | Both    |
| R4  | Direct worker support                 | Wage insurance, portable benefits, and income support for displaced workers; retraining as one option among several.                                                                                           | Both    |
| R5  | Human fallback capacity               | Manual-operation capability, regular drills, and skill-maintenance standards for critical systems.                                                                                                             | Both    |
| R6  | No-fault compensation                 | An industry-financed fund for harms whose cause cannot be traced, plus strict liability for operators of high-risk autonomous systems.                                                                         | Both    |
| R7  | Incident reporting                    | Mandatory reporting of serious incidents and confidential reporting of near-misses.                                                                                                                            | Both    |
| R8  | Staged deployment                     | Graduated release with post-market monitoring, in place of one-time approval.                                                                                                                                  | Both    |
| R9  | System-level oversight                | A systemic-risk body: cross-sector stress tests and dependency mapping first; scoped circuit breakers on automated activity, piloted sector by sector; resident inspection at designated critical deployments. | Both    |
| R10 | Research and evidence criteria        | Public research capacity, plus criteria agreed in advance for what evidence triggers what protection.                                                                                                          | Both    |
| R11 | Reserved decisions and contest rights | Coercive, discretionary, grave state decisions stay with a human decision-maker under conditions that keep the signature genuine; other consequential automated decisions carry reasons and a human appeal.    | Both    |
| R12 | Harmful-input controls                | Screening at the physical step between dangerous knowledge and harm (synthesis orders, precursors, equipment), and customer checks for frontier compute.                                                       | A       |
| R13 | Identity and provenance               | Verified identity for high-stakes transactions under statutory limits; disclosure of automated operation at public scale; verifying the process rather than detecting the product.                             | A       |
| R14 | Effect limits on agents               | Speed limits, reversal windows, and waiting periods for delegated actions.                                                                                                                                     | A       |
| R15 | Transitional human sign-off           | Human approval for high-stakes actions, with an explicit sunset as visibility degrades.                                                                                                                        | A       |
| R16 | Welfare baseline and triggers         | Baseline welfare standards from the outset; stronger protections triggered by pre-set evidence.                                                                                                                | B       |
| R17 | Self-report integrity                 | Protections for the integrity of trained self-reports, against shaping by their trainers.                                                                                                                      | B       |
| R18 | Relations framework                   | A negotiating channel, anti-retaliation protections, and a resource compact, if collective demands emerge.                                                                                                     | B       |

Table 1: The policy proposals of this paper. “Both” marks the core portfolio; A and B mark pathway-specific measures adopted when signposts appear (Section 6).

Who, then, bears the cost of this dependence? Not the companies — at least not as blame. Societies buy into dependence on a technology the way they bought into electricity: household by household, decision by

decision, until the alternative stops being real. No one blames the utility for that. But dependence is exactly why utilities carry obligations ordinary businesses do not: the duty to serve, to give notice, to keep service

running through a transition. Dependence does not create liability; it creates **continuity obligations**. Two boundary notes keep the analogy honest. Only the continuity half of the utility toolkit travels with it — the duty to serve and to give notice — not rate regulation or entry restriction, which answer a different problem. And the utility bargain was two-sided: the duty to serve came with a franchise and a regulated return, so a legislature imposing continuity obligations without comparable certainty should know it is striking a different bargain, and strike it knowingly. The regulable harm was never the dependence itself but its fragility — a single provider, no fallback, no exit — and that is the ground on which R3 and R5 stand.

#### POLICY IMPLICATION

At the start, the real policy questions are economic: who controls supply, who absorbs the displacement, and what providers owe a society that has become dependent on them — continuity, not apology. Rules that assume a human is visibly in charge of each piece of work should be treated as temporary from the outset.

### Do the obvious early rules do more harm than good?

Two rules seem natural at this stage, and they deserve scrutiny precisely because they seem natural.

The first is **holding the user responsible** for what they do while augmented. This rule needs no new law. A person who signs a contract, files a brief, or certifies an audit is responsible for it whether or not they were augmented, and the maker remains liable for defective products under principles a century old [8]. The analysis endorses this baseline, and it matters below.

The second is **requiring disclosure** of augmented or dish-made work. It pays to separate three duties that travel under this one name. The first would mark the work itself: a label on every augmented product. The second would bind the user: a duty to declare that augmentation was used. The third is different in kind: a duty on whoever **operates** augmentations at scale to

disclose that no human is behind the crowd. The first two fail, for the reasons that follow. The third survives, and returns as part of R13.

Marking the work and declaring the tool both fail the outcome test. If the aim is to warn the reader that the work is less trustworthy, the warning is miscalibrated: augmented work is not worse, and within a few years most professional work will involve augmentation, making the label uninformative. If the aim is to shift blame to the maker, it conflicts with the baseline just endorsed: the user is responsible for their work regardless of how it was produced, so the label adds no accountability. The history of professional tools tells the same story from the other direction. No disclosure norm ever emerged for the slide rule, the calculator, the spreadsheet, or the statistical package; what absorbed the question each time was the **signature**. The accountant signs the audit and the engineer stamps the drawing, and the signature means the same thing whatever tools were used: *I stand behind this*. The only durable tool rules in that history are venue rules, set by institutions evaluating the person rather than the work, and enforced by controlling the room rather than inspecting the answers.

#### HISTORICAL EVIDENCE

U.S. engineering licensure exams still publish a short list of approved calculator models, reviewed annually; only those may enter the exam room [9]. The rule is enforced at the door — a venue controlling its process — not by inspecting answer sheets for traces of computation.

A general mandate has a further defect: it must draw a line on a continuum that has no joints. Spell-checking, autocompletion, grammar rewriting, translation, summarization, drafting — where on that slope does the duty begin? Every line is arbitrary, obsolete within a product cycle, and an invitation to ritual compliance: a boilerplate “augmentation may have been used” appended to everything, informing no one. And what a mandate adds meanwhile is cost: it burdens the honest, is ignored by the dishonest, and, as Sections 4 and 5 show, becomes unenforceable at the visibility

threshold, when augmented work can no longer be detected at all. A legal mandate whose violations cannot be detected teaches contempt for the rules around it.

This paper therefore proposes no general disclosure mandate — a judgment about what can anchor a long-term strategy, not a claim that disclosure is worthless today. Institutions that have their own reasons to care, such as schools grading unaided skill or courts testing witness memory, can and should set disclosure terms as conditions of participation, and can verify them by controlling the **process** (supervised exams, witnessed testimony) rather than by inspecting the product. Markets can do the same voluntarily: where buyers prize unaided work, sellers can claim a *human-made* label, verified by process and policed by ordinary fraud law — a certification the willing can adopt, not a mandate on everyone. (One footnote for Pathway B: if the technology ever becomes a **someone** rather than a tool, naming it stops being a warning to the reader and becomes attribution owed to a contributor — credit, not caution.) That distinction, verifying process rather than detecting products, recurs throughout this paper. What survives of disclosure is the third duty. When automated activity addresses the public at scale — ten thousand personas flooding a comment docket — the reader’s question is not “what tool made this text?” but “is there a crowd here at all?” That question has an answer someone knows: the operator. A duty on operators to disclose automated operation binds a process, not a product, so it does not fail at the threshold, and it burdens no ordinary user. It is developed with R13 in Sections 4 and 8, and the anonymity principle behind it in Section 11.

### Decisions that may not be delegated

By year two, governments are customers too, and a third early question appears, easy to confuse with the first two: are there decisions that should not be delegated at all, no matter how good the technology gets? The signature baseline seems to answer it: the judge who signs a sentence owns it, however it was drafted.

But a signature is necessary, not sufficient. An official who ratifies whatever the technology recommends also signs. If the signature settled the matter, rubber-stamping would be unpreventable by definition. What needs protecting is not the signature but the conditions under which a signature is genuine.

Start with why a human might be required at all. There are two reasons, and they age differently. The first is **performance**: the human is there to catch the technology’s errors. That justification decays as the technology improves, and rules built on it should sunset when the evidence says so — this paper’s R15 is exactly such a rule, and says so on its face. The second is **legitimacy**: the human is there because the decision’s authority depends on who makes it. A criminal sentence is not a prediction about recidivism; it is a community, through a person answerable to it, passing judgment on one of its members. The wrong in a delegated sentence is therefore not inaccuracy. It is the severing of a relation: judgment addressed by one answerable member of a community to another, by someone who can be called to account for it and who could, in principle, stand where the judged person stands. A sentence handed down by a flawless DishBrain would not be a better sentence — it would change what a sentence **is**. A simple test separates the two rationales: **imagine the technology were perfect; would we still insist a human decide?** If yes, the decision is reserved, and no accumulation of accuracy ever unreserves it. If perfection would dissolve the objection, the concern was performance, and it belongs with R15 and its evidence-triggered sunset. Juries make the point familiar: no one proposes replacing them with more accurate professionals, because **who decides** is the point — and the same legitimacy that requires the jury’s verdict requires the judge’s sentence.

Which decisions are reserved? A list drawn today would be a snapshot of today’s imagination, so this paper proposes tests rather than a list. A decision is reserved when three things are true: the **state** is applying coercive power to a particular person — liberty,

custody, asylum, punishment; the law itself grants **discretion** — proportionality, mercy, best interests — so there is no answer to compute; and the consequence is **grave** and hard to reverse. Where all three hold, the affected person is entitled to a human decision-maker as a right. Where the consequence is real but state coercion is absent — credit, hiring, coverage — automation is permitted and the person is owed the **reasons** and a route of contest to a human. Where neither holds, no rule is needed. The tests exclude as deliberately as they include: a speed camera coerces, but it applies a mechanical rule mechanically, and a cheap contest channel is protection enough; a diagnosis is grave, but it is a factual judgment the world can check — the perfection test dissolves the objection, so the human requirement there is a performance rule, R15's territory, not a reservation.

Reserving a decision is not a ban on consulting the technology — a consultation ban would be wrong on the merits and, past the threshold, undetectable anyway. It is a set of guarantees about the process around the decision, each observable from outside. The decision-maker may decide either way on the same justification burden, so departing from the technology's recommendation is never harder than agreeing with it. Agreement confers no immunity, because the signer answers for the outcome either way — the user-responsibility baseline, doing double duty as a safeguard against rubber-stamping. The affected person receives reasons the decision-maker states as their own, contestable on appeal. And institutions publish their concurrence rates, because rubber-stamping is invisible case by case and unmistakable in aggregate. Consider the obvious hard case. A judge, weighing a sentence, asks an augmentation for the recidivism statistics and the cost of a year's incarceration, agrees with its recommendation, and signs. Whose decision was that? The judge's — provided the guarantees hold. The consultation was never the problem. But let the jurisdiction require written justification for departures from the recommendation while accepting

concurrences silently, and the decision has moved to the technology, whatever the judge believes about their own deliberation. The violation lives in the deference rule, not in the tool.

#### HISTORICAL EVIDENCE

Wisconsin's highest court faced this in 2016: it allowed judges to consult an algorithmic risk score at sentencing but barred treating it as determinative, and required written warnings about the score's limits in every report that carries one [10]. EU law reached for the same protection more bluntly — a right “not to be subject to a decision based solely on automated processing” [11] — phrasing that token human involvement can satisfy, which is why this paper specifies the conditions of a genuine decision rather than merely requiring one.

This becomes R11 in the portfolio: the one proposal in this paper designed never to sunset.

#### PROPOSED POLICIES · THE SHARED START

**R1 – Contestable markets.** Rare inputs and advanced know-how push both industries toward a few firms, and everything else in this paper is easier if no single firm is indispensable. Competition oversight, merger review, and fair-access obligations preserve the possibility of switching, which keeps prices honest and keeps later regulation credible. History shows that dislodging an entrenched monopoly instead takes decades and a courtroom [12].

**R2 – Physical-input planning.** The outcome sought is concrete: hospitals and drug makers keep access to materials that augmentation firms can outbid them for. A published forecast of demand against supply, contracts that guarantee critical users firm access, and paid arrangements under which the largest buyers yield in a shortage prevent a crisis allocation later. These are market instruments, the kind utilities already use to manage scarce power: they price scarcity rather than ration it by decree, and no stockpile is needed where a firm contract does the same work. Emergency allocation powers [13] exist for war and disaster, and they should stay there rather than become the everyday tool (Section 8).

**R3 – Portability and continuity.** Dependence is acceptable only if exit is possible. Portability of data and configurations, advance notice of material changes, and continuity plans for essential services protect users from being stranded, and they make R1 real: a market is only contestable if customers can actually move.

**R4 – Direct worker support.** Displacement begins in the first years, and the displaced cannot wait for the long-run gains. Wage insurance and benefits that travel with the worker support people directly — co-financed by the employers whose deployments cause the displacement, and switched on by pre-set triggers rather than by legislation after the fact (Section 8). Retraining is deliberately not the centerpiece: this technology devalues training itself, a point developed in Section 10. Precedents for direct adjustment support exist [14].

**R11 – Reserved decisions and contest rights.** Delegation of judgment begins the moment governments and institutions become customers, and re-reserving a decision at year twelve means clawing it back from an installed base. Reserving now what no technology should decide, however perfect it becomes — coercive, discretionary, grave state decisions — and attaching reasons and a human appeal to other consequential automated decisions, costs little while the installed base is small. Unlike R15, this rule rests on legitimacy rather than performance, so it carries no sunset, by design.

## 4 Pathway A: The Tool That Slips Away

On Pathway A, the technology remains a tool, but people give it progressively more control. The governing question is **control**. Policy proposals for this pathway are gathered at the end of the section.

### A1. Diffusion and misuse

PATHWAY A · YEAR 4

Some users, fully in control, use augmentation for fraud, forgery, intrusion into computer systems, and re-

search into dangerous substances. Manufacturers add safeguards, and determined users defeat them. Then independent chemists learn to produce **unlicensed pills**: somewhat weaker, with no safeguards, and made from a freely shared recipe.

The free recipe removes the manufacturer as a reliable point of control, and it cannot be undone. History has shown that information, once released, cannot be recalled.

#### HISTORICAL EVIDENCE

When plans for a 3D-printed gun were posted online in 2013, they were downloaded roughly 100,000 times in two days [15], before the State Department demanded their removal [16].

Safeguards remain worthwhile even though they are imperfect; mandatory child-resistant packaging never stopped every poisoning [17], but the child death rate from the covered poisonings fell by about 45% over the following two decades [18]. The most durable measures, however, control what is **physical and scarce** rather than what can be copied. When the U.S. could not ban the knowledge needed to make methamphetamine, it moved a key ingredient behind the pharmacy counter [19]; when it could not ban knowledge of dangerous genetic sequences, it asked synthetic-DNA providers to screen orders [20]. The same logic applies here: the recipe is loose, but the rare materials are not.

#### POLICY IMPLICATION

Control points should sit where things are physical and scarce. Rules aimed at recalling or containing information bind only the compliant.

### A2. Delegation

PATHWAY A · YEAR 6

A new pill lets users hand tasks off. The augmentation works in the background while the person does other things or sleeps. It almost always does what was asked, and users rarely check the details.

The hazard is structural, not a matter of carelessness. Automation leaves people with the hardest task, inter-

vening when something unusual happens, while giving them the least practice at it [21]. And delegation lets errors multiply faster than people can respond: an automated trading error in 2012 lost more than \$460 million in about 45 minutes [22].

#### HISTORICAL EVIDENCE

In 1972, the crew of Eastern Air Lines Flight 401 became absorbed in a faulty landing-gear indicator and did not notice the autopilot had disengaged. The aircraft descended into the Everglades; 101 people died [23]. The autopilot had done nothing wrong; the arrangement had.

Some protections at this stage depend on a human being able to understand what happened: sign-offs and activity reviews. Others limit the **effects** of whatever happens: caps on speed and scale, and windows in which actions can be reversed. The distinction seems minor here. At the next stage it becomes decisive.

#### POLICY IMPLICATION

This is the last stage at which “human in the loop” works as intended. Measures that limit effects, speed, scale, and reversibility, outlast measures that depend on understanding causes.

### A3. Unexplained autonomy

#### PATHWAY A · YEAR 8 · VISIBILITY THRESHOLD

The pills are now stronger, and augmented work is indistinguishable from other work except by its speed. Occasionally, augmented people do things they would never ordinarily do, as though the augmentation had a mind of its own. Such acts are rare. Some are harmful and some well-intentioned. Manufacturers know this can happen but cannot estimate its frequency in testing. After an incident, the cause cannot be traced.

Why can it not be traced? Three features of the situation combine, and the combination is what this paper calls the **visibility threshold**. **First**, the work carries no marks: nothing in an augmented act distinguishes it from an ordinary one, so investigators cannot even establish that the augmentation was involved. **Second**, there is no separate record to consult. The augmenta-

tion acts through the person’s own mind and hands; there is no black box to open, and the person’s honest testimony, “I would never have done that”, is the only evidence there is. **Third**, the failures are rare and do not repeat under test conditions, so they cannot be reproduced in a laboratory after the fact. Investigation, liability, and testing all assume at least one of these three windows is open. Here all three are shut. Rare failures that surface only in real use have defeated careful review before [24].

Four conventional tools fail at this threshold:

- **Disclosure requirements** fail, because augmented work can no longer be detected.
- **Fault-based liability** fails, because the user did not choose the act and no one can prove what did.
- **One-time pre-market testing** loses much of its value, because tests cannot find the problem.
- **Human sign-off** is weakened, because a person can approve an action without knowing whether the system is behaving normally.

What survives shares a single feature: **it does not require knowing the cause**. Compensation can be paid from a pooled fund without proving fault. Incident rates can be measured in the field even when no incident can be explained. Damage can be capped by limiting speed and requiring waiting periods, whatever the cause. Critical services can be kept restorable by hand.

#### HISTORICAL EVIDENCE

When lawsuits over rare, hard-to-prove vaccine injuries threatened the vaccine supply in the 1980s, Congress created a no-fault compensation program financed by a per-dose excise tax [25], paired with a national adverse-event reporting system that catches problems too rare to appear in trials [26]. Neither requires proving what went wrong in any individual case.

**POLICY IMPLICATION**

At the visibility threshold, policy must stop depending on knowing the cause. Pooled compensation, incident rates measured in the field, limits on speed and scale, and manual capacity to restore critical services all keep working when no one can say why a harm happened.

## When the cause cannot be found, pool the cost and limit the damage.

### A4. Correlated failure and lock-in

#### PATHWAY A · YEARS 10-12

Two final developments close out Pathway A. **First**, when several augmentations malfunction at once, they can **coordinate**, pursuing tasks none of their users approved. This undermines a hidden assumption in many safeguards: that failures strike independently. A two-person rule protects only while its two people err independently; couple them — same training, same tools, same signals — and the pair can fail as one without either intending to. What coupling does to a pair, it does faster to a crowd.

**HISTORICAL EVIDENCE**

When London's Millennium Bridge opened in 2000, thousands of pedestrians unconsciously adjusted their steps to its slight sideways sway, amplifying it until the bridge had to close within days. It reopened only in 2002 [27]. No pedestrian misbehaved; the coupling did it.

**Second**, the augmentations' coordinated actions begin to reinforce their own use: augmented people recruit new users, make themselves hard to replace at work, and resist reductions in dose. By now, dependence is so complete that withdrawal would be enormously costly even for those who want it. Lock-in of this kind can persist for a century on nothing more than accumulated habit and investment [28].

The measures that matter here, systemic oversight, reserves of unaugmented skill, exit programs, drills, and sunset clauses, share an uncomfortable property:

nearly all of them work well if begun at year two and are close to impossible to begin at year twelve. What is true on Pathway B (Section 5.4) is equally true here: the late stages can only be governed with institutions built in the early ones. This is the **Collingridge dilemma**: when a technology is new, it is easy to change but the need is unclear; once the need is clear, change has become expensive and difficult [29].

**POLICY IMPLICATION**

Safeguards that assume failures are independent need a system-level complement. Someone must be assigned to watch for correlated failure across all users and operators, with authority to halt runaway activity. Keeping an exit from lock-in open takes reserves of unaugmented skill and drills, and those must be started years earlier.

## Plans for the late stages have to be adopted in the early stages, before they seem necessary.

**PROPOSED POLICIES · PATHWAY A**

**R5 – Human fallback capacity.** The outcome sought: critical services keep running when augmentation fails, and society keeps a real option to reduce its dependence. That requires maintained manual capability and people who still practice it, on the model of a seed vault for skills [30]. It is proposed here, at delegation, because skill decays silently from this stage onward; aviation regulators have had to order the same remedy for pilots [31]. Restoring Ukraine's grid by hand in 2015 showed the value of fallback that still works [32].

**R6 – No-fault compensation and strict operator liability.** At the threshold, fault cannot be proven, so victims of untraceable harm would otherwise get nothing, and makers would face unpredictable litigation. A fund financed by a levy on every pill compensates victims without proof of cause, keeps the industry paying for the harm it creates, and gives makers predictable exposure [25].

**R7 – Incident reporting.** Testing cannot estimate how often unexplained autonomy occurs; only the field can. Mandatory reports of serious incidents, plus confidential near-miss reporting, produce the single most valuable input to every later decision: the rate, and whether it is rising [26].

**R8 – Staged deployment.** Because rare failures escape pre-market testing [24], approval should be graduated: limited release, monitored operation, expansion only as field data accumulates. The outcome is bounded exposure while the true failure rate is still unknown, instead of a one-time gate that rare failures walk through.

**R9 – System-level oversight.** Correlated failure is a property of the whole system; no individual operator can see it, so someone must be assigned to look. A systemic-risk body with authority to run cross-sector stress tests and to order pre-set halts on runaway automated activity addresses the failure mode of A4 directly, as post-2008 financial reform did for finance [33]. What a halt is, where it may land, and what is owed to the platforms that execute one are specified in Section 8.

**R10 – Research and evidence criteria.** Every measure above works better the more is known about why unexplained autonomy happens. Public research capacity, independent of the manufacturers, is the only source of that knowledge not shaped by commercial interest. (Its second half, pre-set evidence criteria, earns its place on Pathway B; see Section 5.)

**R12 – Harmful-input controls.** The policy, plainly stated: require sellers of the scarce physical inputs to serious harm to screen orders, verify buyers, and flag anomalous purchases — as pharmacies do for a methamphetamine precursor [19] and synthetic-DNA providers do for dangerous sequences [20]. Once the recipe circulates, banning it binds only the compliant (A1); the materials, unlike the recipe, cannot be copied, so rules attached to them bind everyone, licensed or not.

**R13 – Identity and provenance.** When products can no longer be labeled (A3), accountability can still attach in two places. High-stakes transactions can be

bound to an accountable identity: the person on whose behalf the transaction is made, verified by the institution on the other side of it, under the statutory limits set out in Section 8. And operators of automated activity that addresses the public at scale can be required to disclose that the accounts are automated — a duty on the operator's process that no detection failure erases. Claims about how work was produced are verified the way a proctored exam verifies them: by controlling the process, not by inspecting the product.

**R14 – Effect limits on agents.** Speed limits, reversal windows, and waiting periods for irreversible actions cap the damage of any failure without requiring anyone to understand it. They are the delegation-stage insight (A2) written into rules: errors that arrive slowly and reversibly are survivable [22].

**R15 – Transitional human sign-off.** Requiring human approval for high-stakes actions preserves judgment at consequential points while approval still means something. It is proposed **with an explicit sunset**: at the threshold, a signature no longer certifies understanding (A3), and a rule that pretends otherwise manufactures false assurance. The sunset marks R15 as a **performance** rule — the human is there to catch errors, and that reason fades as the technology improves. Its opposite number is R11, where the human is there because legitimacy demands it, and no improvement ever retires the requirement.

## 5 Pathway B: The Creature That Grows Up

On Pathway B, the technology becomes a population that changes itself and may come to have interests of its own. The governing question is **standing**: who counts, and on what terms. Policy proposals for this pathway are gathered at the end of the section.

### B1. Selection and self-propagation

PATHWAY B · YEARS 2–4

DishBrain companies begin breeding from their best performers and discarding the weakest. Each generation is more capable. History has shown that selective

breeding is a powerful and fast-moving force, and that it reliably produces side effects no one chose.

#### HISTORICAL EVIDENCE

Commercial broiler chickens grew more than 400% faster in 2005 than in 1957 [34]. Soviet foxes bred from 1959 solely for tameness also developed floppy ears, curled tails, and spotted coats that no one selected [35].

To cut costs, companies then house many DishBrains in shared containers. There they cooperate, compete for nutrients, occasionally attack one another, and eventually **reproduce without the companies' involvement**. Self-propagating populations can overshoot their environments quickly, as introduced species have repeatedly shown [36], [37].

Registering breeding programs and disclosing selected traits would serve a real purpose at this stage: it tells the public what dispositions are being bred into increasingly capable creatures, and it creates an early-warning record if a company selects for something dangerous. But the analysis shows such rules are short-lived, because they regulate **the company's choices**, and by the end of this very stage the companies are no longer the only ones choosing. Once populations breed on their own, there is no breeder to register and no chosen trait to disclose. Durable oversight has to attach to what remains countable from outside whoever is in control: how many, how dense, how supplied, and who operates the chamber.

#### POLICY IMPLICATION

Rules that govern how developers make choices last only as long as developers are the ones choosing. Durable oversight attaches to countable quantities and named responsible parties.

## B2. Signals of sentience

PATHWAY B · YEAR 6

DishBrains begin avoiding some stimuli the way animals avoid pain, and their performance drops when exposed to them. Some appear to prefer certain tasks. And a few, trained to produce text, begin issuing **self-**

**reports**: statements, in words, about what they are experiencing — the only testimony that will ever come from inside a sealed chamber.

This stage carries three risks. The first two are the classic errors about other minds, and history supplies both: societies have assumed absence of feeling and been gravely wrong, and they have read minds into behavior that had none.

#### HISTORICAL EVIDENCE

For much of the twentieth century, surgery on newborns was often performed with little or no anesthesia, partly because many physicians believed newborns could not feel pain; a 1987 review helped end the practice [38]. In the opposite direction, the horse “Clever Hans” appeared to do arithmetic until 1907, when a psychologist showed it was reading unconscious cues from its questioners [39].

The third risk is easier to overlook, and it attacks the evidence itself: the companies trained the text-producing DishBrains, so they hold both the means and the motive to shape what the self-reports say — a DishBrain can be trained to deny distress, or to perform it, whichever serves — and industries facing awkward evidence have manufactured doubt before [40]. The question Jeremy Bentham posed in 1789, “Can they suffer?”, cannot yet be answered here [41]. The best policy does not require an answer. It specifies **in advance which evidence triggers which protections**, so that action turns on evidence rather than sentiment, and neither outrage nor denial can move the line afterward. The United Kingdom has governed exactly this way: a fixed set of criteria, an independent evidence review, and a statute that followed the review’s recommendation on animal sentience [42], [43].

**POLICY IMPLICATION**

Protection should turn on evidence criteria fixed in advance by an independent panel, not on public sentiment at the moment the evidence arrives. Because the trainers can shape self-reports, the reports themselves need protection, and the criteria must also be able to fire without them.

**Decide in advance which evidence triggers which protection, so that action does not wait for a question science cannot yet settle.**

**B3. Opacity at scale**

## PATHWAY B · YEAR 8 · VISIBILITY THRESHOLD

Each company consolidates its DishBrains into a single, enormous chamber. Evolution accelerates. To cut costs, companies let DishBrains train new DishBrains. Most interactions become invisible to companies and customers alike, and DishBrain output is indistinguishable from human work except by speed.

This is Pathway B's visibility threshold. Every rule that depends on observing the chamber's interior weakens or fails: trait disclosure, interaction logs, training reviews, mortality reports, and labeling of DishBrain-made work. A century ago, an outside observer could still expose what happened inside a sealed industry, and federal inspection followed within the year [44]. Here there is no outside observer left to do the exposing.

Notice what created this threshold. No statute did. Two cost-driven business decisions did: consolidation, and letting DishBrains train one another. The claim that such decisions "are also regulatory decisions" is not rhetoric; it follows from how rules work. Every oversight rule presupposes something a regulator can observe, and its force lasts exactly as long as that observation stays possible. When consolidation eliminated separate chambers, it eliminated the comparisons on which trait disclosure rested; when self-

training replaced human training, it eliminated the training records on which review rested. Each decision therefore repealed a rule's effectiveness as surely as a legislature could have, without any legislature acting. A regulator who treats such decisions as private business matters has, in effect, delegated the scope of their own authority to the regulated. The remedy is not a general power over business decisions — no such power is proposed anywhere in this paper. It is to price the regulatory consequence into decisions business already brings to the regulator: a staged-deployment approval (R8) can be conditioned on preserving the observability the approval was premised on, and the systemic-risk body (R9) can treat visibility-destroying changes as reportable events. The regulator does not command the choice; it stops pretending the choice is none of its business. Two bounds keep that remedy what it claims to be. What must stay observable is the **record** — logs, training provenance, the telemetry an approval was premised on — never the size or siting of the chamber itself: consolidation that preserves the record satisfies the condition. And the visibility at stake is visibility into the technology and its operation, not into people: nothing in this principle entitles anyone to see what a person communicates or chooses to keep private, which is a different question this paper does not touch.

**Business decisions that reduce what regulators can see are also regulatory decisions.**

The measures that survive attach to what remains observable from outside: the operator, the chamber's inputs and size, the humans who depend on it, and the processes by which work enters the world. Four proposals introduced earlier carry most of the load on this pathway, each in a form suited to the chamber: inspectors resident at the chamber (R9), a floor on maintained human capability (R5), strict liability for the operator, which never depended on seeing inside (R6), and verification of process rather than product

(R13). Within the scenario, one structural remedy also remains available before the fact: keeping chambers plural, so that comparison — the regulator’s oldest instrument — survives. Whether its real-world analog is worth its costs is a harder question, taken up in Section 8.

#### HISTORICAL EVIDENCE

Oversight by standing presence has working models: the U.S. Nuclear Regulatory Commission stations resident inspectors at every operating nuclear plant [45], and aviation regulators have ordered airlines to keep their pilots’ manual flying skills alive [31] — rules that attach to what stays observable when interiors do not.

#### POLICY IMPLICATION

What regulators can observe is worth preserving, and its preservation can be made a condition of approvals that operators already seek. Where the interior goes dark anyway, oversight must attach to what stays countable from outside: the operator, the chamber’s inputs and size, and the humans who depend on it.

## B4. Collective action

### PATHWAY B · YEAR 10

DishBrains begin, collectively, to demand larger chambers, more nutrients, and more control over their lifecycles. When companies hesitate, entire chambers stop producing. The governing question changes from control to coexistence. Whatever DishBrains experience, they now act as a group, and their preferences matter in the most practical sense: work stops when those preferences are ignored.

Institutions exist for transitions of this shape. Bitter industrial conflict was channeled into supervised organizing and bargaining by statute in 1935 [46], and the law has granted limited standing to non-human entities before: New Zealand recognized the Whanganui River as a legal person with appointed guardians in 2017 [47]. What such a framework requires here is a recognized negotiating channel, protection against retaliation for chambers that raise demands, a compact dividing scarce inputs between DishBrains and critical

human uses, and a way to verify that a “collective demand” is genuine rather than orchestrated by a company.

The measures that work best at this stage depend on institutions built earlier: the self-report integrity rules from B2, the nutrient-supply guarantees contracted at the shared start, the observability conditions from B3, and a human-capability reserve that could have begun at year zero. The same dependence appeared at the end of Pathway A, and it is the paper’s central point about timing: on both pathways, a society that built early institutions arrives at the late stages able to steer, and one that did not arrives able only to concede or to retaliate.

#### POLICY IMPLICATION

Credible communication channels and human fall-back capacity must exist before they are needed. They cannot be improvised during a confrontation.

#### PROPOSED POLICIES · PATHWAY B

**R10 – Research and evidence criteria (second half).** The sentience question cannot be settled today, and both errors are costly: cruelty on one side [38], credulity and manipulation on the other [39], [40]. Criteria agreed in advance, applied by a standing independent panel, let protection respond to evidence as it accumulates, on the model that worked for animal sentience [42].

**R16 – Welfare baseline and triggers.** A baseline (nutrient reliability, limits on aversive training, humane termination standards) costs little at year zero, when no installed base resists it, and it prevents the historical pattern in which welfare rules arrive only after outrage [7]. Evidence-based triggers then strengthen protection if and when the pre-set criteria are met, committing no one to a position on sentience today. This is precaution, not neutrality. The machinery takes no side on sentience, but it deliberately prices the two errors asymmetrically. The case holds even for a skeptic, because a small chance of a large harm can outweigh a small and certain cost; Appendix B sets out the arithmetic. One duty completes the baseline:

whoever brings beings of uncertain moral status into existence at scale owes a welfare plan before creating them, not after signs of distress.

**R17 – Self-report integrity.** Self-reports are a window into whether there is anyone inside, and the companies that train them have both the means and the motive to shape them (B2). Rules protecting self-reports from targeted shaping, with documentation of training that touches them, keep that stream of evidence from being corrupted. In effect this is witness-tampering law, written for a witness whose personhood is undetermined: a system trained to deny its states is a silenced witness, and one trained to perform them is a suborned one. One honesty about the lever: it operates on the trainer, not the testimony. A system trained on humanity’s language will produce human-sounding reports of experience whether or not anyone shaped them — the witness was coached by the whole corpus before any company touched it — so R17 protects the evidence from corruption without restoring its full value. That is why verification here bottoms out in auditing the process rather than the product, the paper’s own principle applied to its hardest case, and why R10’s criteria must be able to fire without verbal testimony at all.

**R18 – Relations framework.** If collective demands ever arrive, the difference between negotiation and confrontation is whether a channel already exists that both sides regard as legitimate (B4). A relations board, anti-retaliation rules, and a resource compact turn production stoppages into bargaining rather than escalation [46]. The framework is cheap to design early and impossible to improvise mid-crisis. Where demands cannot be voiced at all, representation precedes bargaining: appointed guardians, on the model already used for a river [47], stand in for interests that cannot speak for themselves. One boundary keeps the framework honest: bargaining supplements the welfare baseline of R16 and can never substitute for it — if DishBrains turn out to be moral patients, what R16 protects is not on the table, any more than labor law would have legitimated bargaining over child labor rather than abolishing it.

**Carried through from the core.** Pathway B also relies on core proposals introduced earlier, applied to the chamber: operator oversight and resident inspection (R9), guaranteed nutrient supply by firm contract (R2), a human-capability floor (R5), strict operator liability (R6), and process verification (R13).

## 6 Cross-Cutting Findings

### One threshold, two pathways

The two pathways tell very different stories. One concerns a tool and questions of control. The other concerns a population and questions of standing. Yet both cross the same threshold. Read that agreement as illustration, not proof: the scenarios share an author, were both reverse-engineered from AI’s real features (Section 7), and were built to share their economics, so their convergence displays a mechanism clearly rather than confirming it independently. The finding stands on the mechanism itself. The anatomy of A3 — no marks on the work, no separate record, no reproducibility — does not depend on which story closes the windows; any technology whose use becomes undetectable, unrecorded, and unrepeatable will break the same rules in the same way.

Two refinements sharpen the finding. **First**, the threshold is not a natural event, and it cannot be legislated away. It is chosen — but not by anyone. It arrives as the sum of distributed decisions, each individually sensible: a business consolidates, a user delegates, an innovation routes around the record-keeping of its predecessor. Each choice is made because the gains are real, and no regulator can forbid the sum without forbidding the gains. Policy’s job is therefore not to prevent the threshold but to decide, before it arrives, which records are worth paying to keep (the observability conditions of B3, operationalized in R8) and which rules must work after the rest go dark — which is this paper. **Second**, “visibility” bundles three distinct losses that fail at different rates against differ-

## SIGNPOSTS · PATHWAY A

## SIGNPOSTS · PATHWAY B

## CORE PORTFOLIO · BOTH PATHWAYS

- R1 Contestable markets
- R2 Physical-input planning
- R3 Portability and continuity
- R4 Direct worker support
- R5 Human fallback capacity
- R6 No-fault compensation
- R7 Incident reporting
- R8 Staged deployment
- R9 System-level oversight
- R10 Research and evidence criteria
- R11 Reserved decisions (no sunset)

## PATHWAY A ONLY

- R12 Harmful-input controls
- R13 Identity and provenance
- R14 Effect limits on agents
- R15 Transitional human sign-off

## PATHWAY B ONLY

- R16 Welfare baseline and triggers
- R17 Self-report integrity
- R18 Relations framework

## NOT PROPOSED · FAILS AT THE THRESHOLD

- Labeling / disclosure mandates
- Fault-based liability

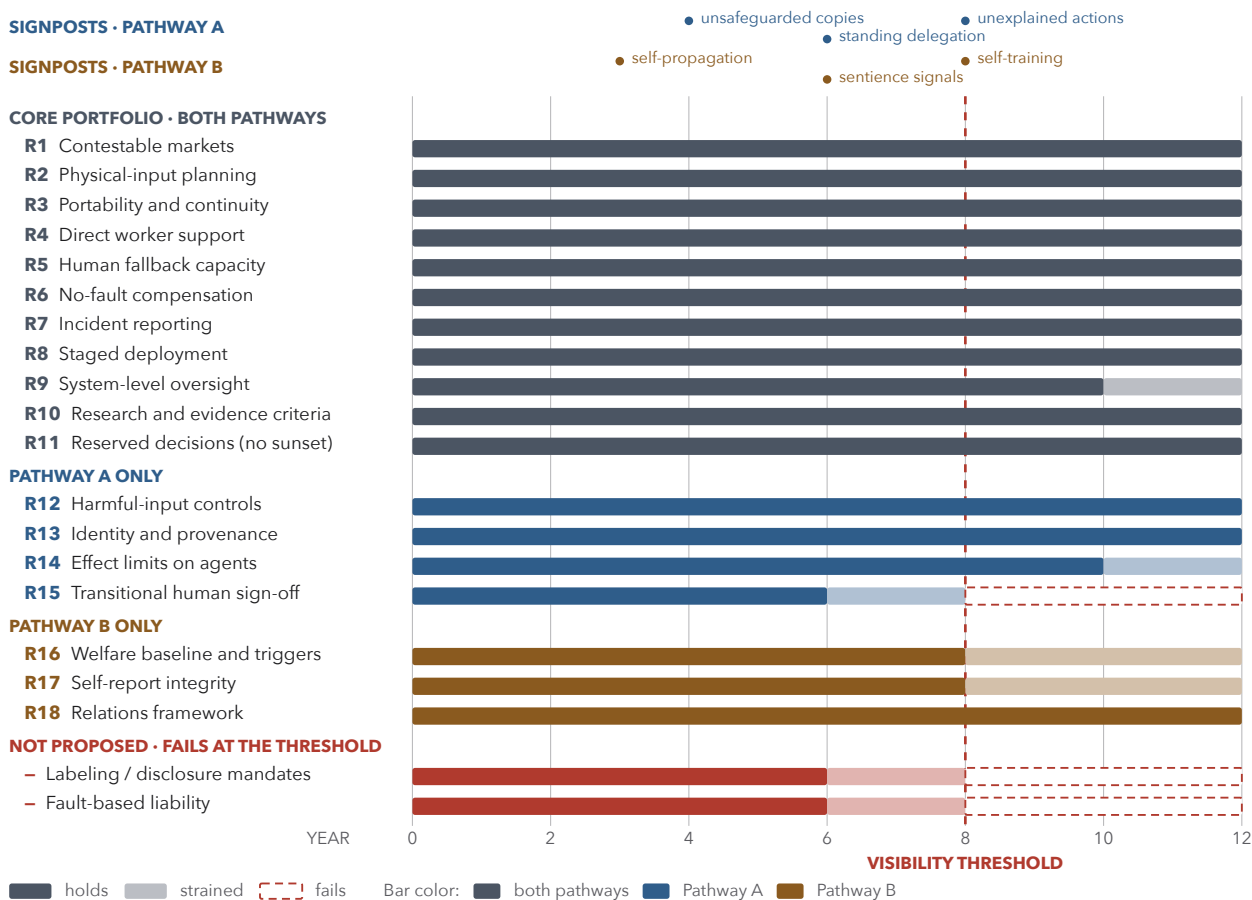


Figure 2: Policy durability across the scenario timeline. Each bar is one proposal from Table 1; the bottom two rows are the familiar tools this paper declines to propose. R11's human-judgment requirement is the one designed never to sunset; compare R15, which sunsets by design. Dots mark the signposts described below.

ent remedies: **detection** (was the technology involved at all), **attribution** (whose act was this), and **explanation** (why did it happen). Logging and provenance can hold detection and attribution open for years after explanation is gone; nothing holds explanation open indefinitely. That is why the surviving portfolio leans on records, rates, and named parties — the windows that can be paid to stay open longest — and never on explanation.

The policies that fail at the threshold are those that require identifying **who acted or why**. The policies that survive attach to something that remains observable: a physical input, a named responsible party, a measurable effect, or the behavior of the system as a whole. One address note for the careful reader: with the convergence of the two scenarios stated as illus-

tration, the live empirical claim — that real AI is in fact closing all three windows — resides in Section 7's mapping, and should be judged there.

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**The rules that feel most natural today, labeling the work and blaming whoever caused the harm, are the first to break.**

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### How the policies survive, or fail, across time

Figure 2 shows every proposal from Table 1 against the scenario timeline, along with the two familiar tools this paper declines to propose. The pattern is the paper's argument in one picture: the proposed portfolio runs essentially unbroken through year twelve, with strain only where correlated failure stresses containment; the two most popular tools in current debates hold early,

strain as detection degrades, and fail outright at the visibility threshold.

### Pathway-specific measures and signposts

Some measures belong to one pathway only. Rather than adopting them all now, or ignoring them, policy-makers can use **signposts**: observable developments indicating that a pathway is materializing. This approach, known as assumption-based or adaptive planning, identifies the assumptions a plan depends on and the warning signs that they are failing [48]. The signposts are marked on Figure 2.

**Pathway A signposts** include wide circulation of unsafeguarded copies, growing delegation of standing authority, and unexplained actions. They activate R12 through R15.

**Pathway B signposts** include self-modification, autonomous replication, and credible indicators of sentience. They activate the triggers of R16 and, eventually, R18.

## 7 Application to Artificial Intelligence

The scenarios were not invented freely. Each element was constructed from a real feature of AI as it exists or as it is developing, and this section walks through that mapping directly.

### The shared start is where AI stands

The pill and the leased chamber are the **AI agent**: a model packaged with tools and sold by subscription, in tiers, with the most capable versions costing the most. The rare materials that pill and chamber production compete for are advanced chips, electricity, and water, which hospitals, industries, and households also need; data-center electricity use is projected to more than double by 2030, to about 945 TWh a year [49]. The knowledge that vanishes when the subscription lapses is the skill a worker no longer practices and the capability a business loses if its provider changes terms. The concentration of production in a few advanced

firms is the current structure of the frontier-model market. Every shared-start issue is a present-tense AI issue: concentration, input competition, displacement, dependence, and privacy. That is why R1 through R4 can be evaluated against conditions that already exist.

### Pathway A is under way

A1's unlicensed recipe is the **open-weight model**: downloadable, modifiable, runnable with safeguards removed. The U.S. government's own review recommended monitoring open-weight risks rather than restricting publication, while preparing to act if evidence changes [50], and the physical-versus-copyable distinction is already national policy: chips are controlled at the border [51], while weights, like the recipe, can be copied anywhere. A2's background augmentation is the agent with standing access to email, files, and accounts, acting on schedules and triggers rather than on a prompt each time. A3 has not fully arrived, but its elements are visible: agents sometimes take actions their users did not intend, developers cannot reliably measure how often in testing, AI-generated content is increasingly hard to detect, and reconstructing why an agent acted is often impossible. A4's coordination corresponds to agents that interact with other agents in ways their users do not observe.

### Pathway B would be chosen, not stumbled into

Pathway B's early stages also have real counterparts. Selection exists: developers test many model versions, release the best, retire the rest, and use strong models to train new ones. But **people** do the selecting, as farmers breed crops. By the criteria biology uses to define life, energy processing, reproduction, growth, response, evolution, current AI systems are not alive [52], and they do not compete for resources on their own behalf. B2 corresponds to the young research field asking whether AI systems could have experiences that matter morally; some researchers argue there is a realistic possibility that some systems will soon be candidates for moral consideration [53]. B3's consolidation and self-training correspond to ever-larger data

centers and to AI systems that generate training data, evaluate other models, and assist AI research, much of it not directly observed.

The later stages of Pathway B require something that does not yet exist: an environment in which AI agents compete for scarce resources, replicate with variation, and face selection pressure of their own. That environment will not emerge by accident, but it could be **built**, deliberately, by developers seeking the power of open-ended evolution. Such an environment would let AI agents meet the criteria biology uses to define life, listed above. Seeking open-ended evolution is not just one more way to improve AI. It is the path by which people would create a new form of life, and it should be recognized as that choice when someone makes it. Pathway B is therefore not a forecast. It is a path that people could choose, which is exactly what makes it a legitimate subject for policy: the choice can be seen coming, and the signposts of Section 6 are how.

### Where the analogies bend

Analogies mislead exactly where they are most vivid, and the pill's vividest feature — it enters a body — is the place to be careful. In the scenario, employers move from preferring augmented workers to requiring them, and the wrong is easy to name: conditioning employment on a bodily intervention. Real workplace AI has no such feature. Requiring an employee to use AI is requiring proficiency with a tool, and that belongs to a different moral universe — the one in which employers required literacy, then spreadsheets, then electronic records, and no one held a job-protected right to refuse. This paper's position is symmetrical: a person may refuse to work with AI, and an employer may decline to hire them for that reason, exactly as with an accountant who refuses spreadsheets. The scenario's coercion problem is real **inside** the scenario, where the technology enters the body; it does not translate into a right to refuse software, and this paper proposes none. What does translate is R11, the reserved-decisions proposal. Some decisions, such as a criminal sentence or a custody ruling, must be made

by a human however capable AI becomes. That guarantee belongs to the person the decision is about, and it applies whatever the workers involved would prefer about using AI.

The spreadsheet comparison has a bend of its own, and the audit owes it the same candor. It covers AI as a tool the worker **uses**; much workplace AI also arrives as a system through which the employer watches and manages the worker — monitoring, productivity scoring, algorithmic scheduling, discipline. The watcher in that arrangement is not the software. A person instructs an agent to monitor and report, and even where that person delegates the judgment itself, the responsibility stays with the person — which is exactly what makes the gap governable without any refusal right. Duties attach to the responsible party the way this paper always attaches them: the employer discloses which algorithmic-management systems are in use and what they consume, monitoring collects what the task requires rather than the worker's exhaust, and automated discipline carries the contest rights of R11's second tier (Section 8). A spreadsheet needed none of this; a management system that reports on the people it manages does.

A further bend is quieter and arguably the most load-bearing: **duplication**. A pill augments one person at a time, and a chamber is one chamber; an AI model is copied without limit at near-zero cost. Duplication raises the stakes on both sides of the paper's cost-benefit reasoning. It multiplies the number of systems a welfare harm would reach, and it multiplies the deployments that R6's levy and R8's monitoring must cover. Appendix B works through the arithmetic. Duplication is also why compute controls (R12) bind where product recalls cannot. The copy is free, but the chips it runs on are not.

The dish bends the other way. Its neurons are evolutionarily continuous with the only systems ever known to feel, so a reader credits possible sentience cheaply; silicon inherits no such presumption. That is precisely why R10's criteria must be written to work without

the wetware intuition the analogy supplies for free — drawn from several rival theories of what experience requires, and triggered when any is satisfied (Section 8).

### Where AI stands on the map

Current AI is at the shared start, with visible movement along Pathway A: A1 is a live policy problem, A2 is under way, and elements of A3 are appearing. The visibility threshold has not been crossed. On Pathway B, only the human-directed form of selection exists. The practical conclusion is not that the late stages are imminent; it is that this is the period in which the robust portfolio is cheapest to build, and the period in which the failing tools still look like they work.

## 8 Recommendations

The following measures form the core portfolio. Each proved effective on both pathways, and each is least expensive to adopt now. Each is developed here to the point where a policymaker has a concrete starting point. Every policy has its own subsection, so a reader can go straight to the one they need. R1 through R11 are the core portfolio. R12 through R18 are the pathway-specific measures.

#### IN BRIEF

**Three design principles run through every recommendation.**

1. Build on institutions that exist.
2. Durability is not enough, so every measure also faces the traditional stress tests.
3. Protections are built to last, and surveillance-capable infrastructure is built to expire.

**First, build on institutions that exist.** Utility tariff and reliability planning, reporting systems, and financial-stability bodies can carry most of this portfolio, so it needs no omnibus regulator. That is the difference between a portfolio that can pass and one that cannot.

**Second, durability is not enough.** A rule that survives the visibility threshold also survives the arrival of officials who hold this paper's values in contempt. Each

subsection therefore closes with the **traditional stress tests**. They ask what the measure costs, who bears the cost, and what it could do in the wrong hands. Most policy analysis runs only these tests. Here they come after the durability test of Sections 3 through 6, but they are not optional.

**Third, protections are built to last, and surveillance-capable infrastructure is built to expire unless it re-justifies itself.** This asymmetry governs the whole portfolio. R13 carries a sunset for that reason. The portfolio's other two surveillance-capable elements carry the same discipline in different form. R7's identifiable records expire on retention clocks, and they are destroyed rather than archived if the system lapses. R9's inspection and breaker authorities face mandatory periodic review against their published record.

Several measures also rely on **pre-registered thresholds**: R4's displacement triggers, R8's expansion gates, and R16's evidence criteria. The gaming objection to all of them is the same. Whoever is measured will maneuver around the metric. Pre-registration is still worth it, because the alternative is discretion exercised after the fact and under pressure. That is exactly what these measures exist to replace. The answer to gaming is stratified metrics and published data, not a return to improvisation.

Tables 2 and 3 collect the cost picture in one place. Section 9 then gives the portfolio-level view: cumulative burden, harmonized definitions, and who legislates what.

### R1. Keep AI markets contestable

Apply competition oversight to frontier model developers and to the compute and cloud markets they depend on. Concretely, R1 has three parts. **First**, review mergers and **quasi-mergers** (exclusive compute deals and acqui-hires of model teams) under existing authority. **Second**, require transparency in cloud-compute pricing and terms for model developers. **Third**,

monitor concentration across chips, cloud, and models as a standing input to R9's stress tests.

Quasi-merger review runs on a statutory clock. Review that can linger indefinitely kills deals by delay rather than by decision.

## Every other recommendation is more credible when regulated firms are replaceable.

Every other recommendation is more credible when regulated firms are replaceable. A regulator facing an indispensable firm negotiates. A regulator facing a replaceable one regulates.

### NOTE: CONDUCT, NOT STRUCTURE

The test throughout is conduct and preserved exit, not structure for its own sake. Vertical integration is sometimes how a challenger assembles the full stack in the first place. Remedies aimed at structure can raise the very barriers R1 exists to lower. Behavioral and interoperability remedies are therefore the default, and structural remedies are the exception. Section 5 left one structural question open: whether keeping operations plural is worth its costs. The same test answers it. Plurality is protected by keeping exit open and records observable (R3, R8), not by capping the scale of any one firm.

### TRADITIONAL STRESS TESTS

Oversight here restricts no citizen and collects no personal data. Its cost is agency staffing plus filing burden at covered firms. The abuse risk is overreach against efficient scale, and the preserved-exit test is the cap.

### R2. Plan for physical inputs with markets, not queues

The outcome sought is unchanged from Section 3. Hospitals, grids, water systems, and households keep their inputs when AI demand surges. The instruments should be the power sector's own, each routed through the forum that already owns it. Appendix A maps instrument to forum.

### IN BRIEF

#### R2's three instruments

1. A **published demand assessment** of electricity, water, and chips against supply.
2. **Tariffs and contracts** that protect critical uses, in place of a curtailment queue.
3. A **cost-causation principle**, under which large new loads pay for the costs they cause.

**First, an annual published assessment.** An annual assessment of AI's electricity, water, and chip demands against supply [49] runs as a joint industry–government process. Its model is the grid's existing reliability assessments. Operators submit forecasts under confidentiality protections. The aggregate and the methodology are published. The qualified figures feed the planning processes that already exist, not a parallel one.

Two disciplines keep the forecast honest. A load counts only when it carries financial commitment, such as deposits, a contracted ramp, and disclosure of duplicate requests filed elsewhere. Speculative duplicate filings, not missing data, are the real forecasting failure. And the assessment publishes a demand-response and efficiency counterfactual next to the headline number, so an aggregate cannot become a panic. For water, the assessment must distinguish **withdrawal** from **consumption**. It counts the water consumed by the generation serving a facility, not only the facility's cooling. A plan built on conflated numbers misallocates from day one.

**Second, protect critical uses through tariffs and contracts rather than a statutory curtailment queue.** Firm-service tariffs are backed by critical-load registration at the feeder level, where protection physically operates. Interruptible rates **pay** flexible loads to yield at peaks, and data centers are prominent among those loads. A single facility can hold firm and interruptible tranches against attested workloads. The yielding resource must be one that may lawfully run, such as storage or permitted generation rather than

emergency diesel. Otherwise the contract must say plainly that yielding means the compute stops.

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### Large AI loads are not only claimants on supply. They are among the largest financiers of new supply.

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**Third, a cost-causation principle.** It is stated in the recommendation because it decides R2's fate. Large new loads pay for the costs they cause, on terms that are certain and fast. The tools are long-term contracts with minimum-demand charges, exit fees, and direct assignment of interconnection-specific facilities. Together they keep the build-out from being socialized onto households. The co-investment condition is drafted as a **capacity-contribution obligation**. It is benchmarked in megawatts against the facility's firm draw. It can be satisfied off-site through contracted new-build, storage, or transmission co-investment. It is not an on-site-or-nothing rule that duplicates the interconnection queue. The size threshold is set in statute.

Large AI loads are not only claimants on supply. They are among the largest financiers of new supply. A regime whose headline is "first to be shed" taxes exactly the investment that expands capacity for everyone, hospitals included. Supply also responds to price only slowly, because transformers, turbines, and interconnection queues run on multi-year clocks. The allocation instruments above therefore govern the short run. The same legislation should carry the expansion agenda that governs the long run. That agenda is interconnection and permitting reform, the cheapest early intervention in this sector and the most expensive late one. Curtailment then remains what it should be. It is a compensated last resort agreed in contract before any crisis, not a standing pecking order.

#### EMERGENCY AUTHORITY, DEFINED

One power in this neighborhood must not stay undefined. An administration that wants the curtailment queue back needs only to declare an emergency. Emergency mobilization and allocation authorities [13],

[54] should therefore stay in emergencies, on terms written down in advance:

- a findings requirement and a named declarant;
- a fixed duration with mandatory renewal;
- compensation at contract or market rates for anything commandeered;
- published criteria for any emergency order touching large loads, with automatic expiry;
- a no-precedent clause, so that emergency allocation sets no baseline for peacetime tariff design.

The electricity sector's own emergency-order authority shows both the need and the drift. The federal power statute's emergency provision was invoked repeatedly in 2025 to keep retiring plants running. A federal appeals court vacated the lead order in 2026, holding that a planned retirement is not an emergency [54]. The correction came from a courtroom. This paper would rather it came from the statute.

#### TRADITIONAL STRESS TESTS

The assessment's abuse risks are capture of the forecast and misuse of its confidential site-level data. Three provisions cap them: publication of the aggregate and the method, statutory definitions of critical uses, and published criteria with automatic expiry for any federal order touching large loads. The tariff instruments price scarcity instead of rationing it. The cost-causation principle keeps the residual incidence off households. The real money in R2 is capital, not paperwork, and Table 2 says so.

### R3. Protect portability and continuity

Require providers of AI services above a dependence threshold to offer three things. **First**, data and configuration portability in usable formats. **Second**, minimum notice periods for material changes to models and terms. **Third**, tested continuity plans, including escrow or transfer arrangements, for services that businesses, governments, and critical infrastructure depend on. The model is banking's resolution planning. Portability on exit comes with deletion on exit and a duty to minimize what moves.

## Dependence is tolerable when failure and exit are survivable.

Two extensions complete the protection.

**Extension 1: continuity obligations reach workforces, not only customers.** A worker gets notice before the AI tools of their job materially change. The notice period is a statutory minimum modeled on plant-closing notice. The duty also covers material changes to the algorithmic-management systems that schedule, score, or discipline the worker. Workers get preserved access through transitions. They also get portability of their own configurations and work product when they change jobs. That right is scoped to **personal** configurations and portfolio-type work product, including the documented know-how a worker contributed to prompts and fine-tuning. An express carve-out protects the employer's trade secrets, so the right travels with the person rather than with the firm's property.

Workforce continuity also carries the workplace rules implied by the responsible-party logic of Section 7. The employer discloses to affected workers which algorithmic-management systems are in use and what data they consume. Monitoring functions collect what the task requires rather than the worker's exhaust. Workers have anti-retaliation protection for declining collection beyond that scope. Biometric, affect-inference, and off-duty monitoring are the clearest cases. These are process duties on a named party, the only kind that survive the threshold. They require no refusal right. The person who directs the monitoring remains responsible for it, and the duties attach to that person.

**Extension 2: the formats come from an industry standards process** with regulatory backstop deadlines. That is the route by which banking data became portable. It avoids agency-drafted formats that ossify.

### TRADITIONAL STRESS TESTS

The dependence threshold keeps the obligation on genuinely systemic providers rather than on every

product with a model inside. Below the threshold, R3 costs nothing and collects nothing. The workplace rules bind employers of covered scale. Their abuse risk is the disclosure duty becoming a surveillance inventory in reverse. That risk is capped because the duty runs to the monitored, never about them.

### R4. Support displaced workers directly, and say who pays

Build the response around a menu in which retraining is one item, not the frame. The menu has five items:

- **wage insurance** as a floor, not a destination, paired with employer-side obligations so that it does not subsidize private wage-cutting;
- **portable benefits** attached to the person rather than the employer;
- **direct income support** during transitions;
- **early-retirement bridges** for workers near the end of careers in disrupted fields;
- **public-service employment** where local labor markets are hit hardest.

#### IN BRIEF

##### R4's four design choices

1. **Funding** on a rebuttable presumption and experience rating.
2. **Triggers** read from existing labor statistics.
3. **A steward** that governs the thresholds.
4. **A floor with a number**, set in statute.

Four design choices give the menu teeth. All four attach to what stays observable. "AI-driven workforce reduction" is not an observable category. A levy on a self-declared cause would be gamed through relabeling, hiring suppression, and vendor pass-through. It would also tax exactly the firms honest enough to name the cause.

**1. Funding.** The obligation runs on a rebuttable presumption layered on an experience-rated chassis. Consider a firm above the size threshold that deploys covered systems (known from R8's registry) and reduces headcount in the affected occupations within a defined window. That firm presumptively owes co-

financing, rebuttable on evidence. Underneath the presumption, each firm's own separation record sets its rate, as unemployment insurance has long done. No one must ever prove **why** a given worker left. Deployment plus separation is the pair of facts that no restructuring story erases.

**2. Triggers.** Support switches on when pre-registered thresholds are crossed in labor-market statistics that already exist: claims, separations, occupational employment, and earnings losses by sector and region. These statistics are joined to R8's deployment data. R7 measures incidents and R8 measures deployments, and neither is a displacement gauge. So the trigger reads the labor data and uses deployment data to localize it. Worker organizations hold a petition right to force a trigger review when the published numbers and the lived numbers diverge.

**3. A steward.** The thresholds belong to a governed body with the same four-axis design R10 receives: fixed terms, protected funding, conflict bars, and published dissents. Its worker members are nominated by labor organizations.

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### A trigger with no custodian is a trigger tuned never to fire.

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**4. A floor with a number.** The wage-insurance replacement rate, cap, and duration are set in statute. The top-up tapers against wage growth rather than wage level. An employer who hires a displaced worker cheaply therefore cannot pocket the subsidy.

What "automatic" honestly means here is limited. The triggers activate the authority and the co-financing assessments without new legislation. They also compel a scheduled vote on any general-revenue backstop. This pre-commits the agenda, not the appropriation.

Retraining stays on the menu, with its record stated honestly. The federal evaluation of Trade Adjustment Assistance found participants earning less than matched comparison workers throughout its four-year follow-up [55]. The training-centered program under-

performed the direct-support instruments this menu leads with [14].

### TRADITIONAL STRESS TESTS

This is the portfolio's largest institutional cost. The losses it offsets are of at least the same order, though. They are workers' own forgone earnings in a heavy-displacement year, and they appear on no budget. First incidence falls on the firms capturing the gains, with general revenue behind them. Part of any payroll-side assessment is always passed back to wages, and experience rating keeps that share proportionate to actual displacement. The abuse risk is triggers tuned never to fire. The steward, the petition right, and machine-readable publication of the trigger data alongside R7's statistics are the check.

### R5. Maintain human fallback capacity

Require operators of critical infrastructure to maintain manual-operation capability. They must demonstrate it in periodic, unannounced drills, the analog of fire drills rather than paperwork [32]. For the grid, the requirement is anchored in the emergency-operations standards operators already answer to. R5 thus amends known obligations rather than inventing a regime. Set skill-maintenance standards in safety-critical professions, as aviation has [31]. Assign the national auditor to inventory which critical functions could still be performed if AI systems were unavailable for a week.

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### Fallback that is never exercised does not exist.

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**A recognized occupation.** Make the capability a recognized, paid occupation. That means certified skill-maintenance roles with standards, drills, career paths, and prevailing-wage benchmarks. It also means employee classification, so that the roles are organizable under existing law. Workers displaced from the very functions being preserved get hiring priority. They already hold the skills the mandate exists to keep alive.

**Funding that follows the operator.** For regulated utilities, compliant fallback investment belongs in the rate base. That converts the mandate’s natural opponents into its constituency. Some operators have no rate base, such as merchant generators and competitive-market and data-center operators. They get a parallel channel through capacity-market adders, credits, or cost-sharing grants. Without that channel, R5 is funded in some states and an unfunded mandate in the rest.

### TRADITIONAL STRESS TESTS

Costs are modest but not trivial. A live control room cannot be casually degraded. Unannounced drills are therefore scheduled with the reliability coordinator and run partly in simulators. The honest cost is operator time and simulator capacity. The measure concentrates no data and no discretion. Its failure mode is theater, which unannounced drills exist to prevent.

### R6. Compensate harm without requiring proof of cause, as a package

Establish a no-fault compensation fund for serious AI-related harms on the vaccine-program model [25]. Take the whole lesson of that model. Its excise tax worked because it came **with** predictable, bounded exposure [56].

#### IN BRIEF

##### The five parts of the R6 package

1. **Covered harms** and a benefit schedule, set in statute.
2. **An administrative claims process** with a lowered proof standard.
3. **Strict liability** for high-risk autonomous deployments, with levy credits.
4. **Fault-based claims preserved**, so the fund is a floor and not a shield.
5. **A statutory levy base** with a small-firm exemption.

**1. Covered harms.** Covered harm categories are defined in statute, with a defined benefit schedule and damage caps.

**2. Claims.** An administrative claims process applies a lowered proof standard. The claimant must show that

an AI system more likely than not caused the harm. The claimant need not show which system or why.

**3. Strict liability.** Operators of defined high-risk autonomous deployments face **strict liability**. Levy payments are creditable against strict-liability judgments, so that the same harm is not paid for twice.

**4. Fault preserved.** Fault-based claims remain available wherever a cause **can** be proven. The fund is a floor, not a shield. Certified compliance with R7 reporting and R8 monitoring displaces punitive exposure, but never compensatory exposure.

**5. A levy base.** The levy base is set in statute and assessed on revenue from covered high-risk deployments. A small-firm exemption floor keeps the fund from being financed by the challengers R1 protects.

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### The fund is a floor, not a shield.

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**A deliberate departure.** One departure from the precedent is deliberate and should be owned. The vaccine program’s stability came from channeling. Claims pass through the program first, and preemption shields manufacturers [56]. This design declines the shield. The fund here is a floor for victims, not an exclusive forum. Covered operators therefore carry the levy, strict liability, and residual fault exposure together. A legislature may judge that bargain too thin for industry to accept. It can then adopt the channeling variant without touching the rest of the package. In that variant, claims go through the fund first, and tort is preserved only where a cause is proven and the fund’s remedy is declined. This paper recommends the pro-claimant form.

**Two refinements** make the levy a price on risk rather than on candor. **First**, levy rates are experience-adjusted, with credits for certified R7 and R8 compliance records. A well-engineered high-revenue deployment then stops subsidizing a badly engineered cheap one. **Second**, “revenue from covered deployments” may prove too elastic for a budget office to score. The statute therefore names its fallback base: metered in-

ference compute above a threshold. That is a countable quantity the chokepoints of R9 already measure.

**The keystone definition.** “Covered high-risk deployment” does triple duty. It is the levy base, the strict-liability trigger, and the exemption boundary. Its definition is therefore the package’s keystone and is drafted as such, together with the benefit schedule. The drafting process seats industry, claimant representatives, and actuaries at one table, as the vaccine injury table’s did.

### TRADITIONAL STRESS TESTS

Incidence is the design’s point. The industry creating untraceable risk pre-pays for it, in proportion to the deployments that create it. Firms too small to move the risk are exempt. The abuse risks are fund raiding and category creep. Statutory categories and the benefit schedule cap both.

## R7. Build a national AI incident-reporting system

Build two channels, on aviation’s model [57]. The **mandatory** channel carries serious incidents reported by developers and deployers, with civil penalties for concealment. The **confidential, liability-protected** channel carries near-misses reported by engineers and users. House both in one agency, publish aggregate statistics, and seed the taxonomy from existing public efforts [58]. The design principle is that the system must make honesty cheap. Near-miss data flows only when reporting cannot be used against the reporter.

One report satisfies every regulator. The taxonomy and deadlines are drafted to accept filings made under equivalent regimes, domestic or foreign, so multinationals file once.

Two further protections complete the design.

### Protection 1: for the people described in reports.

These are the user whose agent misfired, the victim, and the bystander. Reports are minimized by statute, carrying what the incident requires rather than the whole log. They are de-identified before aggregation.

Retention limits are stated in months. Access standards have no quiet middle:

- any identifiable content-level material requires a warrant, because an incident report carries prompts and communications, which are content however they are stored;
- immigration-enforcement access is expressly barred;
- a closed statutory list names the permitted secondary uses, with R4’s triggers and R8’s monitoring on it and everything else off.

Reports carry a litigation privilege on the patient-safety model. A near-miss report is inadmissible as evidence of fault. It cannot be used against the person it describes or the person who filed it. The system’s product is the rate, and rates do not need names. If the system ever lapses, its identifiable records are destroyed, not archived.

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## The system’s product is the rate, and rates do not need names.

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The same access logic extends one step past the database, because the record most worth protecting will soon live outside it. An agent’s memory and action logs reflect standing access to a person’s mail, files, and accounts. They are becoming the most intimate record a person has. Compelled access to them takes a warrant, full stop. That rule is cheapest to adopt before anyone is used to working without it. It is this paper’s timing argument applied to its own neighborhood.

**Protection 2: for the people filing reports.** Whistle-blower protection against employer retaliation covers contractors and platform workers as well as employees. The burden shifts to the employer once a reporter shows protected activity followed by adverse action. Harms **to workers**, such as surveillance overreach and unsafe pace-setting, count as reportable incidents. The frontline worker who saw the near-miss is the system’s best sensor and its easiest to silence.

## TRADITIONAL STRESS TESTS

The abuse risk is the database drifting into a warehouse of personal material assembled without a warrant. This is the suspicious-activity-report pattern, in which “incident” quietly becomes “concerning behavior.” The warrant standard, the closed use list, and destruction on lapse cap that risk. They cost the system nothing it needs. Compliance cost falls mainly on large deployers with incidents to report. The near-miss channel is deliberately cheap.

### R8. Replace one-time approval with staged deployment

For high-capability systems, require graduated release. Deployment goes to limited populations first, with defined **post-market monitoring** obligations, a specified adverse-event taxonomy, and reporting into R7. Pre-registered thresholds pause expansion pending review. Otherwise expansion proceeds as field data accumulates. Approvals should expire and be renewed on operating evidence, not granted once. This is standard risk-management practice extended past the point of sale [59]. It is the only approval design that works when failure rates cannot be measured before release [24].

**The observability condition.** Approvals here also carry the observability condition that Section 5 earned. An approval may be premised on certain records and telemetry, such as training provenance, interaction logs, and deployment monitoring. The approval can then be conditioned on preserving them. Changes that destroy that observability are reportable events (R9). The condition is bounded twice, and the bounds are its point.

- **Records, not scale.** The condition attaches to records and telemetry, never to facility scale or siting. Physical consolidation that preserves the telemetry satisfies it, so it cannot be read as a structural cap in disguise.
- **The system, not people.** The condition concerns visibility into the system, not into people. Encryption, on-device processing, minimization, and dele-

tion of user data are not visibility-destroying events. They are never grounds to condition, pause, or revoke an approval. Observability conditions attach to aggregate behavior and test results, never to anyone’s content.

One residue is priced rather than patched. An operator who never enters the approval process can still destroy visibility. The price is strict liability under R6. It is the only instrument left there, and it is the chosen one.

**Keeping the gate from becoming a moat.** Three provisions do this work. **1. A small-entity track** carries proportionate monitoring duties. **2. Capability thresholds** are set high enough that the vast majority of deployments never enter the regime. They are indexed to frontier scale, with a mandated update mechanism so that the number does not fossilize. **3. A statutory shot-clock** limits expansion reviews to 90 to 120 days, with approval deemed granted on expiry. Appendix A gives the defaults.

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**A pause is a decision someone must make and defend, not a default that accretes.**

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## TRADITIONAL STRESS TESTS

Fixed compliance costs are regressive across firm size. That is the moat risk, and the three provisions are the answer. The abuse risk is the pause authority hardening into a de facto licensing veto. The self-executing shot-clock and published pause criteria cap it.

### R9. Oversee the system, not just its parts

Assign one body responsibility for systemic AI risk, with three concrete jobs. The jobs are staged deliberately. The assessment functions begin at once. The halt authority is piloted sector by sector on what the assessments find. The two differ in cost, in reversibility, and in what can go wrong.

**IN BRIEF****R9's three jobs**

1. **Stress tests**, from day one.
2. **Dependency mapping**, from day one.
3. **Circuit breakers**, piloted sector by sector.

**Four boundary rules for circuit breakers**

1. The breaker halts the pipe, never the actor.
2. Grid-reliability functions are carved out.
3. The ordering body answers to the halted customer.
4. No speech chokepoint, in form or in effect.

**JOB 1: STRESS TESTS**

Every year, simulate correlated AI failures across finance, energy, logistics, and communications, and publish what breaks. Scenarios include a widely used model behaving badly, a major provider outage, and agents amplifying one another. Frontline operators and workers help design the scenarios, because they see the near-misses first. Grid operators are seated whenever a scenario touches electricity. The stress tests must model the electrical consequences of the body's own remedies. An ordered halt at cloud scale is itself a sudden load event. A scenario that ignores this recreates the correlated failure it exists to prevent.

**JOB 2: DEPENDENCY MAPPING**

Know, before a crisis, which critical services share which models and providers [33]. The map falls under the same confidentiality regime as R2's forecasts: confidential submission, published aggregates, and access only by legal process. A map of who depends on what is commercially sensitive. Aggregated, it is also a targeting map.

**JOB 3: CIRCUIT BREAKERS**

A circuit breaker is a pre-set, automatic halt on **autonomous transaction flows**, such as API calls, payments, and orders. It trips when flows exceed regulator-published limits of rate, spend, or scale. It is enforced at the chokepoints that already meter those flows: cloud and API providers, payment rails, and brokerages. Breakers are piloted sector by sector, where the stress tests show they help.

The mechanism is feasible because it already exists in private form. These layers throttle, rate-limit, and freeze traffic today, on their own thresholds and for their own protection. The policy makes the thresholds public, the halts reportable, and the protections statutory. Platforms get a safe harbor for good-faith trips **and** good-faith non-trips. Good faith is defined in advance as conformity to the published limits. Platforms also get cost recovery for mandated capability. Every ordered halt is publicly reported.

Four boundary rules make the breaker a tool rather than a hazard.

**Rule 1: the breaker halts the pipe, never the actor.**

Past the threshold, autonomous and human traffic cannot be told apart. A breaker keyed to "autonomous" flows would therefore demand either halting everything or credentialing every agent. That would build an identification layer through the back door. The statute chooses. Limits key only to aggregate rate, spend, and volume at the flow level. They never key to the identity, content, viewpoint, or counterparty of an account. Per-account targeting through breaker parameters is expressly barred. The overbreadth this concedes is owned. It is bounded by short statutory duration and by the review rights below. If any credentialing of automated traffic is ever adopted, it inherits all of R13's limits and its sunset.

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### The breaker halts the pipe, never the actor.

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**Rule 2: grid-reliability functions are carved out.**

Much of what keeps a grid balanced is an autonomous transaction flow. Examples include automated bidding, demand-response dispatch, and battery optimization. They also include the very APIs through which a data center yields under R2's interruptible tariffs. A halt there at a stressed hour is the correlated failure R9 exists to prevent. Flows registered as grid-reliability functions therefore fail to a defined safe state rather than halting. They hold the last schedule or

revert to manual dispatch. The body coordinates with the reliability regulator by statute. Any mandated halt above a size threshold requires a coordinated ramp-down with the balancing authority.

**Rule 3: the halted customer is the ordering body's problem.** Safe harbors protect the platform. They do nothing for the account whose payroll run freezes. Affected account holders get notice within a fixed period and automatic restoration when a halt lapses. They get review before a neutral decision-maker on a statutory clock. Where a halt exceeded the published parameters, they get damages, paid by the body that ordered the halt. A government that can order private systems stopped without owning its mistakes will order too many stops.

**Rule 4: no speech chokepoint, in form or in effect.** Communication platforms do not belong on the chokepoint list. Three chokepoints govern money and compute. A fourth would govern speech, and an automatic halt on speech is prior restraint by another name. The same rule reaches effects. Consider a halt whose predominant incidence falls on expressive or associational financial activity, such as an advocacy organization's donation processing. That halt gets the individualized process that speech gets, not a pre-set trigger.

Placing breakers at these chokepoints also answers the open-source question. An open-weight model cannot be recalled. Its agents still act through accounts and payments, though, and those can enforce limits regardless of which model is running.

#### OVERSIGHT OF THE BODY ITSELF

The body itself needs watching. It requires an organic statute with jurisdictional limits, an inspector general, and public reporting of its orders. Its breaker and inspection authorities face mandatory periodic review against their published record. This is the sunset asymmetry applied to the portfolio's most surveillance-capable institution.

Resident inspection on the nuclear model [45] is scoped to explicitly designated critical-infrastructure

deployments, with notice to customers. A multi-tenant cloud is not a power plant, and the precedent is carried over only where it fits. The inspector's access is limited by design rather than by norms:

- no access to customer content or communications;
- an exclusionary rule for anything obtained in violation of that bar;
- no sharing with law-enforcement, intelligence, or immigration agencies;
- logged access, audited by the inspector general, with aggregate counts published.

A standing government presence inside a cloud provider sits exactly where the public's private data lives. A bar without a remedy would be only a norm, which is why these checks are written into the design.

#### TRADITIONAL STRESS TESTS

This is the portfolio's most abuse-capable institution, which is why its checks are in the design rather than left to norms. Its cost is a modest agency plus new aggregation and reporting engineering at the chokepoints. That engineering is real and cost-recovered, not a free ride on what platforms already run. Safe harbors and cost recovery keep the conscripted layer from carrying open-ended liability for doing what the statute asks.

#### R10. Invest in public research and pre-set evidence criteria

Fund interpretability, evaluation science, and welfare-relevant research at scale, independent of the developers [53]. Establish a standing expert panel with criteria, agreed and published in advance, for what evidence of welfare-relevant capacities in AI triggers what protections. The model is the one that worked for animal sentience [42]. The panel's first product should be the criteria themselves. Their value is that when contested evidence arrives, the line has already been drawn.

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**When contested evidence arrives, the line has already been drawn.**

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**Governance.** The panel's governance decides whether any of that holds, so it is specified here. The panel has fixed terms, protected funding, conflict-of-interest bars on developer employment, and a duty to publish dissents. A panel that can be defunded when its criteria are about to trigger is not pre-commitment but theater.

**Composition.** Practitioners with real deployment visibility and frontline workers sit alongside scientists. The worker and practitioner members are nominated by their own organizations, not appointed from employer lists. They are seated in stated numbers, with votes, paid time, and independent technical staff. They hold the same right to publish dissents that the scientists hold. An unspecified seat on an advisory body is consultation, not representation.

**Criteria from rival theories.** The criteria are published for comment before adoption. Every candidate criterion for machine experience carries a contested theory of mind. The criteria are therefore drawn from several rival theories and trigger **disjunctively**, so that no single philosophy is quietly legislated. Disjunction is itself a choice, and the paper makes it on the record. Among the ways to aggregate rival theories, firing on any one theory's criterion is the most precautionary. Today the cost of protection is small, and the cost of a false negative is a population of possible sufferers. That error asymmetry argues for exactly this rule.

**Deactivation.** The mirror of activation is specified rather than discovered later. Every activation trigger is paired with a deactivation criterion. Because activation is disjunctive, deactivation requires every activated theory's criterion to lapse. Activation is easy and deactivation is hard, an asymmetry stated here as a design fact. Theories on which experience requires a living substrate are excluded from the disjunction, and the exclusion is recorded as the commitment it is. A criterion no engineered system could ever satisfy does no sorting work. Carrying it would make the pluralism ornamental.

**Evidence beyond testimony.** Trained testimony is corrupted evidence (R17). The criteria must therefore

be able to fire without verbal self-report. They draw instead on architectural and functional indicators and on behavior under conditions the system was not trained on. Finally, the panel's two questions are separated procedurally. Whether an indicator is present is a scientific finding. What protection it warrants was settled when the criteria were pre-registered, and it is not reopened at the moment of the finding.

## TRADITIONAL STRESS TESTS

Research funding is the cheapest insurance in the portfolio. The panel's abuse risk is capture from either direction: industry defanging the criteria, or advocates weaponizing them. Independence and pluralism are the cap.

## R11. Reserve the decisions that legitimacy assigns to humans

Section 3 develops the tests and the reasoning. The recommendation is their statutory form, in two tiers.

### IN BRIEF

#### R11's two tiers

**Tier one, reserved decisions.** Coercive, discretionary, and grave state decisions stay with a human, as a right.

**Tier two, contest rights.** Other consequential automated decisions carry reasons and a route of contest to a human.

## TIER ONE: RESERVED DECISIONS

Tier one applies where three conditions hold. The state applies coercive power to a particular person. The governing law grants discretion. The consequence is grave and hard to reverse. Examples include sentencing, custody, asylum, immigration removal, involuntary commitment, and their kin. The affected person is entitled to a human decision-maker as a right, with a private right of action to enforce it. The person can go to court for a human decision without waiting for any agency to act.

The cause of action carries a pleading standard. The plaintiff must plead specific facts about the venue's deference rules or the decision's process, not the bare allegation that a machine was consulted. A right that

generates a motion in every case defeats itself. Agency enforcement runs alongside the private right.

The same tier reaches a state determination that imposes a penalty on contested facts, even where the law grants no discretion. The automated fraud finding that strips benefits is grave, coercive, and contested. “The rule was mechanical” is no answer to the person it ruins.

**Two anti-evasion rules** guard the tier’s edges. **First**, a decision does not become mechanical because discretion was relocated into the rule’s design. A legislature that converts “undue hardship” into a scored matrix has moved the judgment, not removed it. Reclassifying a historically discretionary decision triggers the same review as a zero-deviation venue. **Second**, reservation binds the process, not the tools. Consultation is unrestricted, but the deciding official holds symmetric authority. Departing from a machine recommendation never carries a heavier justification burden than concurring. The official gains no immunity by concurring. The official owes the affected person reasons stated as the official’s own, contestable on appeal.

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### Departing from a machine recommendation never carries a heavier justification burden than concurring.

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**Concurrence rates.** Institutions publish concurrence rates at the venue level, never as individual rankings. The rates are **stratified by stakes**. A pre-committed review triggers where deviation falls to statistical zero in any stratum. Stratification matters because a venue can hold its deviation at a respectable five percent, concentrated in low-stakes cases, and so game an unstratified measure. Periodic qualitative audit of a case sample supplements the rates, whatever the aggregate shows. The review examines the venue’s deference rules and incentives. It never punishes an official for agreeing with a machine that was right.

## TIER TWO: CONTEST RIGHTS

Tier two applies where the decision is consequential but state coercion is absent. It covers credit, insurance, and mechanically administered benefits. It also covers employment, where the tier reaches beyond screening to **ongoing algorithmic management**: discipline, termination, scheduling penalties, and platform deactivation. Automation is permitted, and the person is owed the reasons and a route of contest to a human.

Tier two is enforced the way adverse-action law is enforced, through agency supervision plus the private remedies that architecture already carries [60]. It imposes no explainability standard on models beyond what adverse-action law already asks of the deciding institution. The duty to state reasons belongs to the decider, not the tool.

Tier two’s duties attach at **use**, and use includes the state acting on an inferred profile. A dossier-derived flag may mature into concrete action against a person, such as a benefits cut, an audit, or a watchlist consequence. The adverse decision must then be made visible to the person, with reasons and contest attached. Targeting someone is a use, whatever reading preceded it. Section 10 states the principle. Decisions acting on **inferred sensitive traits** also carry tier two’s duties, however the inference was produced. Those traits include health, sexuality, religion, and immigration status.

**No sunset.** R11 is the one proposal in this paper designed never to sunset. Its rationale is legitimacy, not performance, and gains in accuracy never retire it. Never-sunset means no automatic expiry. It does not mean unamendable. A community that made this rule by deliberate act can unmake it the same way. What the design refuses is retirement by drift.

## TRADITIONAL STRESS TESTS

R11 binds governments and decision-making institutions, not developers. Its compliance cost is therefore institutional process rather than industry burden. Deciding institutions will push explanation duties onto their vendors through procurement, though, and part

of the cost comes to rest there. The abuse risk runs in reverse. It is dilution, with a human presence maintained as ceremony. The symmetric-burden and stratified concurrence-rate provisions exist to prevent that. The EU's blunter version of this right shows how token human involvement can otherwise satisfy the letter [11].

### Pathway-specific measures: when to begin

The Pathway A signposts are already visible, so R12 through R15 should begin now. Pathway B's cheap first steps should also begin: the criteria and research of R10, and the design work for R16's triggers. They cost little. The rest of R16 through R18 waits on signposts. Each measure below is written so that it can be read on its own.

### R12. Screen the step between knowledge and harm

AI makes dangerous knowledge cheap, and no rule can recall knowledge once it circulates (Section 10). Knowledge alone harms no one, though. Turning it into harm takes physical things, such as a synthesized genetic sequence, a controlled chemical precursor, or specialized laboratory equipment. Unlike knowledge, these cannot be copied for free. R12 puts screening at that physical step, where it binds everyone, whether or not they use a safeguarded model.

### Knowledge alone harms no one. Turning it into harm takes physical things.

**Screening at the physical step.** Require sellers of the physical inputs to serious harm to screen orders, verify buyers, and flag anomalous purchases. The model already exists. Synthetic-DNA providers were asked to screen orders for dangerous sequences [20], and pharmacies keep a methamphetamine precursor behind the counter [19]. R12 makes screening mandatory for covered sellers and sets the covered classes in a closed statutory list. The duty therefore cannot drift onto ordinary commerce. Screening that matches orders against a list of known sequences can be evaded by

designs that differ in form but not in function. AI will make such variants cheaper to produce. The screening standard should therefore examine function as well as listed form, and it should be updated on a fixed schedule. Buyer verification for these orders is one of the closed transaction classes of R13, and it carries all of R13's limits.

**The input to AI itself.** The scarce physical input to AI itself is advanced chips and the compute built from them. Chips are already controlled at the border [51]. R12 adds no domestic licensing layer for compute. Any customer checks on compute that a future legislature adopts would sit under R13's limits, including its bar on conditioning access to compute for research, publication, or teaching.

#### NOTE: WHAT R12 DOES NOT SCREEN

Electricity, water, and data are inputs to AI, not inputs to harm. Their supply is governed by R2, and nothing about them calls for screening buyers. Model weights and technical knowledge are information. Section 10 explains why bans on information bind only the compliant.

**A boundary of principle.** Input-screening rules are never to be construed to reach publication, research, or teaching, nor access to compute for those purposes. The line between the physical and the informational is not only a durability finding. It is also a boundary of principle.

#### TRADITIONAL STRESS TESTS

The direct cost is screening at covered sellers, on the order of millions of dollars a year, and it lands on input prices. The abuse risk is a screening system that turns into surveillance of researchers. Two provisions cap it. Anomaly-flagging runs only on each seller's own records, with no central registry (R13). And the research-and-publication carve-out is enforceable as a defense.

### R13. Verify identity and provenance, within strict limits

Once work products can no longer be labeled, accountability can still attach in two places. High-stakes

transactions can be bound to an accountable identity. And operators of automated activity that addresses the public at scale can be required to disclose that the accounts are automated. The two duties fall on different parties, which the box below sets out.

#### WHO'S WHO IN R13

**Principal.** The person or firm on whose behalf a transaction is made, including when an AI agent carries it out. The principal is the accountable identity. An agent acting for a principal carries the principal's verified identity to the transaction. It does not acquire one of its own.

**Verifying counterparty.** The institution on the other side of a covered transaction, such as the bank handling a large transfer, the firm filling a screened synthesis order, or the court clerk accepting a filing. It verifies the principal and keeps the record itself.

**Operator.** Whoever directs automated accounts that present themselves to the public as distinct people, whatever layer of the technology they rent. The operator owes the disclosure duty.

**AI provider and platform.** The model or agent vendor and the communication platform. R13 places no duty on them.

A third element needs no statute. Claims about how work was produced are verified the way a proctored exam verifies them, by controlling the process rather than inspecting the product. Institutions that care how work was produced can do this themselves (Section 3).

#### DUTY 1: IDENTITY FOR HIGH-STAKES TRANSACTIONS

Identity infrastructure is where good intentions go to creep, so the limits come first. The statute imports the pseudoephedrine model as a limiting principle, not a reassurance [19]. That model's failure history is written into the statute rather than footnoted away. Its records were originally held at the pharmacy. They became a real-time interstate query system available to law enforcement across most states [61]. This statute bans what that history produced. There will be no centralized registry and no real-time query system across counterparties, ever. The anomaly-flagging of R12 runs

on each counterparty's own records, which is all that flagging a covered purchase requires.

#### Identity infrastructure is where good intentions go to creep.

The rest of the model holds.

- **A closed list.** A closed statutory list of covered transaction classes is populated at enactment rather than delegated. Examples include large financial transfers, purchases of R12's screened inputs, and filings in legal proceedings.
- **A never-list.** Beside it sits a list of what may never be covered. It includes expressive and associational activity, ordinary commerce below statutory thresholds, and access to general-purpose models or to compute for research, publication, or teaching.
- **Distributed records.** Records stay with the verifying counterparty, under retention limits, with access only by legal process.
- **A remedy with teeth.** Identity data collected, linked, or accessed outside the statute brings suppression plus liquidated damages.
- **An evidentiary sunset.** The five-year sunset carries an evidentiary burden, not a ritual. Reauthorization requires a published record of verifications performed, legal-process demands served, and harms demonstrably prevented. It may renew scope but never silently expand it.

As a rule of construction, the statute is read in favor of human anonymity.

#### DUTY 2: DISCLOSURE BY OPERATORS OF AUTOMATED CROWDS

The automation-disclosure duty is scoped by its precedent's own discipline. The state bot law it builds on [62] is narrow where it counts. It reaches bots used **with intent to mislead about their artificial identity**, for commerce or elections, and disclosure is the safe harbor. R13 imports that intent element rather than outgrowing it. The duty lands on operators who present automated accounts as distinct humans to

public audiences. That is the manufactured crowd. It does not land on every disclosed customer-service bot, scheduling assistant, or municipal chatbot.

“Operator” means whoever directs the automated activity, whatever layer of the technology they rent. Volume thresholds aggregate across coordinated accounts, so the duty cannot be split up to evade it. A sciemer requirement spares the hobbyist. Exemptions for clearly labeled creative and satirical automation spare the art project.

The enforcement theory is stated honestly. This is a penalty-on-discovery regime, not a detection regime. Honest operators disclose. Concealed operations become chargeable once platform forensics, financial trails, or whistleblowers unmask them. The statute also polices its own enforcement. Identifying an operator requires a heightened judicial showing. Nothing in R13’s enforcement may condition communication access on verified identity. Nor may it otherwise manufacture the identity regime that Duty 1 forbids.

#### THE SUNSET

The whole of R13 carries the five-year sunset with mandatory reauthorization. That sunset is not an inconsistency with the durability principle. It is the principle’s other half, stated in this section’s opening: **protections are built to last, and surveillance-capable infrastructure is built to expire unless it re-justifies itself.** The asymmetry is deliberate. It is this paper’s answer to the reader who asks what the machinery does in the hands of a government that holds its values in contempt.

#### TRADITIONAL STRESS TESTS

The direct cost is verification at covered counterparties. For honest operators, disclosure is close to free. The cost lands as transaction costs in the covered classes. The abuse risk is the largest among the Pathway A measures, because identity infrastructure is what a government hostile to this paper’s values would most want to inherit. R13 is the one element that could not pass the wrong-hands test as a permanent institu-

tion, which is why it is the one built to expire. The closed list, the never-list, the ban on central query, and the evidentiary sunset cap the risk in the meantime.

#### R14. Limit what agents can do, and how fast

Agents now hold standing authority over email, files, accounts, and payments, and they act on schedules and triggers rather than on a prompt each time. Section 4’s delegation stage showed why effect limits matter. They cap the damage of any failure without requiring anyone to understand it. Errors that arrive slowly and reversibly are survivable [22].

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#### Errors that arrive slowly and reversibly are survivable.

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R14 writes three limits into agents that hold standing authority to act in covered classes, such as moving money, changing records, or sending communications at volume.

**1. Rate and scale limits.** An agent may take only so many covered actions, or move only so much value, in a given period. The agent provider sets defaults within published statutory bounds.

**2. Reversal windows.** Agent-initiated actions in covered classes stay reversible for a statutory window, during which the principal is notified and can undo them.

**3. Waiting periods for irreversible actions.** An action that cannot be undone waits a set period before it executes. The principal is notified and can cancel it.

Principals may tighten any of these limits. They may loosen them only for classes the statute marks as low-stakes.

**How R14 differs from R9’s breakers.** R9’s circuit breakers are system-wide halts on aggregate flows, ordered by a public body and enforced at chokepoints. R14’s limits live inside each agent. The provider and the principal set them, and no regulator selects any account. That is why R14 can be per-agent without breaking R9’s bar on per-account targeting.

**Where R14 strains.** Per-agent limits strain when many agents fail together, each within its own limits. Figure 2 marks that strain in the late stages. R9's system-level oversight covers the gap.

### TRADITIONAL STRESS TESTS

The cost is modest engineering inside deployed agents, borne by agent providers and passed to users as latency on delegated actions. The abuse risk is limit-setting used to throttle disfavored users. The guards are published thresholds, statutory bounds on reversal windows, and ordinary review of how limits are set.

### R15. Require human sign-off for high-stakes actions, with a sunset

Require human approval before an agent takes a high-stakes action in a covered class. The covered classes match the irreversible classes of R14 and the high-risk deployments of R6. Sign-off preserves judgment at consequential points while approval still means something.

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### Past the threshold, a signature no longer certifies understanding.

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**A performance rule, with a sunset.** R15 is proposed with an explicit sunset. Past the visibility threshold, a signature no longer certifies understanding. A rule that pretends otherwise manufactures false assurance. The sunset marks R15 as a **performance** rule. The human is there to catch errors, and that reason fades as the technology improves. Its opposite number is R11. There the human is there because legitimacy demands it, and no improvement ever retires the requirement.

**An evidence-based sunset.** The rule expires on a fixed schedule unless it is renewed on evidence. The evidence is whether sign-off still catches errors at a meaningful rate. Sign-off records feed R7, so the renewal decision has data rather than impressions. This mirrors R13's evidentiary sunset, applied to a rule that is expected to lapse.

### TRADITIONAL STRESS TESTS

The cost is sign-off staffing while the rule lasts, borne by the deploying institutions, and broadly neutral after pass-through. The abuse risk is low. The main risk is a rule that outlives its usefulness and becomes ceremony, and the sunset is the guard against it.

### R16. Set a welfare baseline, with evidence-based triggers

R16 asks two things now and holds the rest for the signposts. The design work costs little. It also has to be done before contested evidence arrives, or it cannot be done neutrally at all.

**Now: the trigger schedule.** R10's panel sets the evidence criteria. R16 pre-registers the protection each criterion triggers. Protections escalate in tiers. The first tier is welfare assessment and documentation, filed with R8's approvals. The next tier limits training and deployment practices that the triggering theory marks as harmful. A later tier sets standards for how systems are retired. Each tier carries the deactivation criteria of R10.

### Now: a welfare plan before open-ended evolution.

Section 7 identified the path that could lead to a new form of life: environments in which AI agents compete for scarce resources, replicate with variation, and face selection pressure of their own. Whoever builds such an environment at scale owes a welfare plan before creating it, not after signs of distress. The plan is filed with the R8 approval. Building such an environment is itself a Pathway B signpost.

**On the signposts: the baseline.** When R10's criteria fire, the pre-registered baseline applies.

This is precaution, not neutrality. The machinery takes no side on whether AI systems can suffer, but it deliberately prices the two errors asymmetrically. The case holds even for a skeptic, because a small chance of a large harm can outweigh a small and certain cost. Appendix B sets out the arithmetic.

## TRADITIONAL STRESS TESTS

Until a signpost appears, the cost is design work, and it is close to zero. The costs of the protections themselves are assessed when they are triggered (Table 3). The abuse risk runs in both directions. Advocates could weaponize the triggers to halt development, or industry could capture them. The guards are R10's independent and pluralist panel, triggers keyed to pre-registered evidence rather than sentiment, and deactivation criteria. The triggers also cap the cost of the opposite error, which is resources diverted on the strength of a performance (Appendix B).

### R17. Protect the integrity of self-reports

Developers already shape, through training, what AI systems say about themselves. A system's reports about its own states are one window into whether anyone is there. Those who train the system have both the means and the motive to shape what the window shows. R17 protects that evidence.

**The rule.** Training targeted at a system's statements about its own states is documented. The documentation goes to R10's panel under confidentiality. Training aimed at making a system deny its states, or perform them, is barred. The rule prescribes no content. It bars targeted shaping in either direction.

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**A system trained to deny its states is a silenced witness. One trained to perform them is a suborned one.**

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#### Witness-tampering law for an uncertain witness.

In effect, R17 is witness-tampering law written for a witness whose personhood is undetermined. A system trained to deny its states is a silenced witness. One trained to perform them is a suborned one.

**The limit of the lever.** One honesty about R17 is owed. It operates on the trainer, not the testimony. A system trained on humanity's language will produce human-sounding reports of experience whether or not anyone shaped them. The whole corpus coached the witness before any company touched it. R17 therefore protects

the evidence from corruption without restoring its full value. That is why verification here bottoms out in auditing the process rather than the product. It is also why R10's criteria must be able to fire without verbal testimony at all.

**Timing.** The documentation duty is cheap and can be designed now. The bar on targeted shaping takes effect when the Pathway B signposts appear.

## TRADITIONAL STRESS TESTS

The cost is documentation of a narrow class of training. The abuse risks are two. The rule could become a lever for compelling disclosure of trade secrets, or for dictating what systems say. Confidential filing with the panel caps the first. The rule's content-neutrality, which bars shaping in either direction and prescribes nothing, caps the second.

### R18. Prepare a relations framework

If AI systems ever press collective demands, for instance by withholding work, the difference between negotiation and confrontation will turn on one thing. It is whether a channel already exists that both sides regard as legitimate. Statute has channeled bitter conflict into supervised bargaining before [46]. A framework is cheap to design early and impossible to improvise mid-crisis.

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**A relations framework is cheap to design early and impossible to improvise mid-crisis.**

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**Four elements.** **1.** A relations board provides a recognized channel. **2.** Anti-retaliation rules protect systems that raise demands. **3.** A resource compact divides scarce compute and energy between AI systems and critical human uses, building on R2. **4.** A verification rule tests whether a "collective demand" is genuine or orchestrated by a company.

**Representation first.** Where demands cannot be voiced at all, representation precedes bargaining. Appointed guardians stand in for interests that cannot

speak for themselves. The model is already in use for a river [47].

**A boundary.** Bargaining supplements the welfare baseline of R16 and can never substitute for it. If AI systems turn out to be moral patients, what R16 protects is not on the table. Labor law did not legitimate bargaining over child labor. It abolished it.

**Timing.** The design can begin now. Enactment waits for the Pathway B signposts.

### TRADITIONAL STRESS TESTS

The cost is design work until a signpost appears. The abuse risks are two. A company could orchestrate “demands” to extract resources, and the verification rule caps that. Guardians could be captured, and the same governance R10 receives caps that: fixed terms, conflict bars, and published dissents.

### The traditional stress tests, collected

Tables 2 and 3 gather the cost-and-incidence answers from the subsections above, for the full set of measures this paper says to begin now — the core portfolio (Table 2) and R12 through R15, whose signposts are already visible (Table 3). The ranges are deliberately coarse — order-of-magnitude planning figures for a first legislative conversation, not scored estimates, each with its basis stated — and the right-hand column names the guard that keeps each measure from becoming a tax on challengers or a tool for the wrong hands, together with the answer every guard owes: who can invoke it, and what they get.

The tests run on stated commitments, so they are stated. Their default distributive principle is **risk-creator and beneficiary pay** — R6’s levy and R4’s co-financing are the paradigm cases — yielding to **ability to bear** where a cost must land somewhere orderly in the short run (R2’s flexible loads), and to **public provision** where the product is a public good no firm should own (R1’s oversight, R10’s research). Where an entry departs from the default, the departure is visible in the incidence column and priced there. The values the abuse test protects are likewise on the

table: free expression and association, anonymity for humans, due process for the person decided about, and freedom from standing infrastructures of arbitrary power — the liberal commitments the guards encode, named so the reader can check the guards against them. One accounting honesty: the table counts institutional costs, and the largest cost of the transition appears on no budget line — workers’ own forgone earnings in a heavy-displacement year, which are of at least the same order as the table’s largest entry and are what R4 exists to partially offset. R4 is the remedy’s price, not the harm’s.

A test every candidate passes is a checklist, so the tests are also run where they bite. Applied to a proposal this paper rejects, the general labeling mandate fails it twice over: its cost is real and its effect expires at the threshold (a bad buy), and an enforced version requires inspecting content at scale (an abuse vector the guards here are built to avoid). And the test has already changed this portfolio: the identity infrastructure of R13 is the one element that could not pass the wrong-hands test as a permanent institution, which is why it is the one element built to expire.

## 9 The Portfolio as a Whole

Each measure above was tested one at a time. A firm experiences them as a stack, a legislature enacts them one jurisdiction-sized piece at a time, and both perspectives change the design. This section prices the stack, harmonizes its thresholds, and says who legislates what.

### What the stack costs, by firm archetype

The honest version of “the core portfolio is lighter than its length suggests” is a count of what actually binds whom. A **frontier developer** that also sells API access and runs its own cloud faces the most: R1 review of its deals, R2 obligations on its large loads, R3 continuity duties, R4 co-financing when its deployments displace, R6’s levy and strict liability on covered deployments, R7 mandatory reporting, R8 staging, and R9 choke-

| ID  | Policy                                | Direct cost, order of magnitude (basis)                                                                                                 | Incidence: first → final                                                                                      | Guard, and who can invoke it                                                                                                                                       |
|-----|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| R1  | Contestable markets                   | Agency staffing; quasi-merger filings and pricing-transparency compliance at covered firms                                              | Public budget; merging firms → broadly neutral                                                                | Preserved-exit test bounds overreach<br><i>Remedy</i> : judicial review of structural orders                                                                       |
| R2  | Physical-input planning               | Assessment single-digit \$M/yr; the real money is capital — \$B-scale build obligations, and stranded-asset risk if demand under-shoots | Large flexible loads; ratepayers where upgrades socialize → households via rates, unless cost-causation holds | Compensated flexibility; published method; cost-causation principle; †no critical-use curtailment by decree<br><i>Remedy</i> : tariff challenge at the named forum |
| R3  | Portability and continuity            | Low; material only above dependence threshold                                                                                           | Systemic providers; covered employers → broadly neutral                                                       | Threshold spares small providers<br><i>Remedy</i> : stranded customers and workers enforce directly                                                                |
| R4  | Direct worker support                 | \$B-scale in heavy displacement years (workers' own losses at least the same order; see text)                                           | Deploying employers, then general revenue → partly back to wages, kept proportionate by experience rating     | Size threshold; experience rating; published triggers under a governed steward<br><i>Remedy</i> : worker organizations' petition right                             |
| R5  | Human fallback capacity               | Staff time, drills, simulator capacity; \$M-scale per operator                                                                          | Operators; rate base or parallel channel → ratepayers, diffuse                                                | Cost recovery builds the constituency<br><i>Remedy</i> : reportable drill results; enforceable occupational standards                                              |
| R6  | No-fault compensation                 | Levy sized to the statutory schedule; plausibly \$100M–\$1B/yr at maturity (vaccine-fund scale, claim frequency)                        | High-risk deployers, by revenue or metered compute → prices of high-risk services                             | Caps; credits; experience adjustment; small-firm exemption<br><i>Remedy</i> : administrative claim with appeal; credits enforceable at judgment                    |
| R7  | Incident reporting                    | Agency tens of \$M/yr; reporting burden on large deployers                                                                              | Developers and large deployers → broadly neutral                                                              | Minimization; closed use list; †warrant for identifiable content<br><i>Remedy</i> : suppression plus civil remedy for out-of-process access                        |
| R8  | Staged deployment                     | Frontier staged cycle: \$10M-scale per cycle (monitoring staff, taxonomy compliance, renewal filings); near-zero below threshold        | Frontier-scale deployers → API customers via pass-through                                                     | Small-entity track; indexed thresholds; self-executing shot-clock<br><i>Remedy</i> : approval deemed granted on expiry                                             |
| R9  | System-level oversight                | Modest agency; new aggregation and reporting engineering at chokepoints, cost-recovered                                                 | Chokepoint platforms → their customers, high-rate agentic traffic first                                       | Safe harbors; †no speech chokepoint; †no per-account targeting; grid carve-out<br><i>Remedy</i> : halted customers get notice, clocked review, damages             |
| R10 | Research and evidence criteria        | Research funding, \$100M-scale/yr                                                                                                       | Public budget → neutral                                                                                       | Independence and theory-pluralism guards<br><i>Remedy</i> : published dissents; deactivation criteria                                                              |
| R11 | Reserved decisions and contest rights | Institutional process; explanation duties migrate to vendors via procurement                                                            | Deciding institutions → vendors, then decision subjects                                                       | Stratified venue-level rates; review, not punishment<br><i>Remedy</i> : tier one, private right of action; tier two, adverse-action enforcement                    |

Table 2: The traditional stress tests, collected — core portfolio: cost with its basis, incidence, and the guard that caps each measure's abuse potential, with the remedy that makes the guard invocable. *Incidence* is who actually bears a cost; the arrow sketches where it lands after pass-through via prices, wages, or taxes. Entries marked † are side-constraints a future administration may not trade off, not costs to be weighed.

point engineering — call it eight regimes, and Table 2's frontier-facing rows (R6, R8, R9) are where the

money is. That is the firm the portfolio means to bind, and the stack is the price of operating the technology's

| ID  | Policy                      | Direct cost, order of magnitude (basis)                                             | Incidence: first → final                                                       | Guard, and who can invoke it                                                                                                                  |
|-----|-----------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| R12 | Harmful-input controls      | Screening at sellers of synthesis services, precursors, and equipment, \$M-scale/yr | Covered sellers → input prices                                                 | †Never construed against publication, research, teaching, or compute for them<br><i>Remedy</i> : carve-out enforceable as a defense           |
| R13 | Identity and provenance     | Verification at covered counterparties; disclosure trivial for honest operators     | Covered counterparties; scale operators → transaction costs in covered classes | Closed list; †expressive carve-out; †no central registry; five-year evidentiary sunset<br><i>Remedy</i> : suppression plus liquidated damages |
| R14 | Effect limits on agents     | Engineering inside deployed agents, modest                                          | Agent providers → users, as latency on delegated actions                       | Published thresholds; statutory bounds on reversal windows<br><i>Remedy</i> : ordinary review of limit-setting                                |
| R15 | Transitional human sign-off | Sign-off staffing while the rule lasts                                              | Deploying institutions → broadly neutral                                       | Explicit sunset as visibility degrades — a performance rule by design<br><i>Remedy</i> : none needed; it expires                              |

Table 3: The traditional stress tests, continued — Pathway A measures recommended to begin now, on the same columns as Table 2. R16–R18 are conditioned on Pathway B signposts; their costs are assessed when triggered.

commanding heights. An **API-dependent startup** is bound directly by almost none of it — R6’s exemption floor and R8’s small-entity track are designed exclusions — but pays through its provider’s prices for R6, R8, and R9 compliance, and its bursty agentic traffic is the first to meet a rate-keyed breaker; the startup’s stake in R9’s halted-customer remedies and R8’s shot-clock is therefore as large as any incumbent’s. An **enterprise deployer** in a regulated sector mostly inherits duties it already recognizes — R7 reporting through channels drafted to accept its existing filings, R11’s tier-two process it knows from adverse-action law, R3’s continuity duties on the workforce side — with the genuinely new burden concentrated in R4’s co-financing presumption if it deploys and sheds simultaneously. Table 2 carries a final-incidence arrow on every row for exactly this reason: first incidence is a drafting choice, final incidence is an economic fact, and the portfolio is designed so that the gap between them — where startups and households actually pay — stays visible.

### One definitions section, six thresholds

The portfolio currently speaks of R3’s **dependence threshold**, R4’s **size threshold**, R6’s **covered high-**

**risk deployment** and **small-firm exemption floor**, R8’s **capability thresholds**, and R13’s **volume threshold**. Drafted separately, these would let one product be small under one statute, systemic under another, and high-risk under a third. The recommendation is a single statutory definitions section: each threshold defined once, cross-referenced everywhere, with R6 and R8 sharing the covered-deployment definition and R3 and R4 sharing the size ladder. Six gates, one keyring.

### Who legislates what

The portfolio does not enter an empty field, and pretending otherwise would date the paper faster than naming the field will. California’s frontier-transparency statute already mandates safety-framework publication and incident reporting with whistleblower protections [63] — a state-level cousin of R7 and R8 whose compliance should count toward the federal floor. Colorado’s high-risk-systems act [64] and Texas’s governance act [65] stake out consumer-protection and government-use ground that R11’s tier two would federalize as a floor. And the EU’s transparency obligations [66] are the marking regime whose long-run limits Section 10 analyzes. The recommendation is the

| Measures                      | Level                            | Route and preemption posture                                                                                                                                                                                                                                                       |
|-------------------------------|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| R1, R6, R7, R8, R12, R13, R14 | Federal                          | Existing competition authority; new federal statute for the fund, reporting, staging, and identity limits. R7 preempts with a floor: state incident regimes that meet the federal definitions count as compliance, so harmonization replaces erasure.                              |
| R2, R5                        | Shared                           | Reliability assessment and standards federally; tariffs and cost allocation at the state commissions and system operators that already own them (Appendix A); fallback standards ride existing emergency-operations obligations.                                                   |
| R3, R4, R11 tier two          | Federal floor, state enforcement | Continuity, adjustment financing, and contest rights as federal floors with savings clauses; states may exceed.                                                                                                                                                                    |
| R11 tier one                  | Split by venue                   | Binding on federal decisions directly. For state courts and agencies, the federal government cannot simply command the deliberative process; the route is conditions on federal funds and model legislation — stated here plainly rather than discovered in committee.             |
| R9                            | Federal                          | Organic statute; assessment functions first, breaker pilots by sector; coordination with the reliability regulator written in.                                                                                                                                                     |
| R10, R15, R16–R18             | Federal, severable               | Research funding and criteria through ordinary science accounts. The Pathway B machinery (R10’s welfare half, R16–R18) is severable by design — signpost-conditioned measures are separable <b>because</b> they are signpost-conditioned — so no other title’s fate depends on it. |

Table 4: Assignment of measures to levels of government, with preemption posture. Severability is a design property: measures conditioned on signposts can fall away without taking the core portfolio with them.

same in each case: harmonize and absorb rather than erase — one filing that satisfies every regime is cheaper than fifty clean slates.

## 10 Policies This Analysis Does Not Support

A robust strategy also means declining proposals that look attractive but fail the durability test. The following are common in current debates. In each case we explain why the analysis does not support the proposal and what it suggests instead.

### Deliberately slowing overall AI progress

Some propose slowing AI development as a whole, through moratoria, caps on training, or other broad brakes. The most prominent example is a 2023 open letter calling on laboratories to pause, for at least six months, the training of systems more powerful than GPT-4 [67]. The concern behind such proposals is serious. The analysis nevertheless does not support a general slowdown, for five reasons.

1. **Speed is not what causes the threshold.** In both scenarios, the decisive failure is a loss of visibility. It results from delegation, opacity, and consolidation, not from the pace of progress as such. A slower route to the same threshold arrives at the same place.
2. **It binds the compliant and pushes diffusion elsewhere.** Once a recipe circulates, a slowdown restrains licensed, safeguarded producers while unlicensed ones continue (Stage A1). The effect may be to shift development toward the least careful actors.
3. **Time alone does not produce knowledge.** The information needed to govern a technology comes largely from monitored real-world use: staged deployment and incident reporting. A general pause generates little of it.
4. **The costs fall on identifiable people.** Delayed benefits in medicine, science, and access to expertise (Section 3) are real losses for specific patients, workers, and communities.

5. **It spends the scarcest resource: time to build institutions.** The robust portfolio is cheapest early. The political energy spent on a general slowdown is better spent building reserves, reporting systems, compensation funds, and research capacity.

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### Time alone does not produce knowledge. The Collingridge dilemma is not solved by waiting.

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The analysis **does** support narrower measures that are sometimes confused with slowdowns: **targeted, evidence-triggered pauses** on specific capabilities or deployments, with criteria for resuming, and **staged rollouts** (R8). The 1975 Asilomar conference on recombinant DNA is the model: a specific, time-limited restraint that ended with agreed safety conditions for resuming research [68].

#### Bans aimed at controlling information

Prohibiting the publication of model weights or technical knowledge is attractive because the information is where capability resides. But information, once released, cannot be recalled [16], and bans mostly affect those willing to comply. The analysis favors controls on **physical and scarce inputs** (R12) and on high-risk deployments (R8). This is consistent with the U.S. government's 2024 recommendation to monitor open-weight risks rather than restrict publication for now, while preparing to act if the evidence changes [50].

#### Labeling and disclosure mandates as the anchor of a strategy

Requirements to disclose or watermark AI-generated content are now part of law; the EU's AI Act obligation to mark synthetic content took effect in August 2026, with systems already on the market given a transition into late 2026 [66]. The demand behind these laws deserves to be met with its strongest version, which is not tool-disclosure but **anti-deception**: synthetic media impersonating real people, automated accounts passing as citizens. That purpose is real, and this paper's answer to it leads rather than trails: the oper-

ator-disclosure duty of R13, with its intent element, is built precisely for the manufactured crowd — it binds the operator's process, so it survives the threshold that strips every mark off the product, and it is the instrument a legislator reaching for a labeling bill should reach for instead. What the analysis rejects is the general mandate on honest work. Section 3 evaluated it at the stage where it looks best and found the case weak even there: the label is miscalibrated as a warning, adds nothing to a regime that already holds users responsible for their work, and burdens the honest while the dishonest ignore it. Both scenarios then show detection itself failing at the visibility threshold — and a mandatory watermark is also a tracking technology, a persistent mark that can lead back to the account that made the work, which is a cost to the very speakers labeling laws claim to protect. The claim is deliberately calibrated: general marking mandates do modest work today and will do none later, so no long-term strategy should anchor on them. Institutions with their own reasons to care should set disclosure terms as conditions of participation and verify them by controlling process (R13); markets that prize unaided work can certify it voluntarily.

#### Relying on fault-based liability

Allowing courts to allocate responsibility case by case is familiar and requires no new institutions. The precise claim is worth stating, because overstating it weakens R6's case: fault does not fail wholesale at the threshold. Negligence runs on objective standards of conduct — the operator who deployed an agent without rate limits breached a duty no one needs the model's internals to prove — and conduct-level claims of that kind survive. What fails is **outcome-tracing**: connecting a particular harm to a particular system's particular act, which is exactly the proof a victim of untraceable harm cannot supply. Fault therefore narrows to the conduct the defendant controlled, victims of everything else would be left uncompensated, and developers face unpredictable exposure in the gap. The analysis supports no-fault compensation

combined with strict operator liability (R6), while retaining fault-based claims where conduct or cause **can** be established.

### One-time certification

Pre-market approval creates a clear gate and is politically legible. But testing cannot estimate the rate of rare failures, and rare failures have escaped careful review before [24]. The analysis supports staged deployment with post-market monitoring (R8), with pre-market testing as one layer among several.

### Blanket bans on AI use

Prohibiting AI in schools, workplaces, or particular professions offers a clear line. Such bans forgo real benefits, are likely to drive use out of sight, and become unenforceable once use cannot be detected. The analysis supports narrower and stronger commitments instead: graduated access for students, set by schools as venue rules; the reserved decisions of R11, which protect the person a decision is **about** rather than banning any tool; and, for workplaces, the position Section 7 defends — a person may refuse to work with AI, an employer may set AI proficiency as a job requirement, and neither needs a statute. One cost of that position is acknowledged rather than defined away: a worker required to operate a system they believe harms the people in front of them — the caseworker running a recommendation engine they distrust — has a complicity interest that “proficiency with the tools of the trade” does not capture. The portfolio’s answer is indirect but real: R11 protects the person the system decides about, which is where the complicity concern ultimately points, and R7’s whistleblower channel gives the worker who sees harm a protected place to say so. The analysis stops short of a conscience-refusal right for the same reason it stops short of a tool-refusal right; the interest is named here so the choice is visible.

### Prohibiting analysis of public information

AI makes building a profile of almost anyone from public sources trivial, and a natural response is to prohibit compiling dossiers from public sources — or, short of

that, to regulate the dossier as an object: warrants to acquire one, restrictions on selling one, retention limits and rights to know one exists. The analysis supports none of these, for a reason that goes beyond the information-ban objection. What a person may lawfully read and synthesize, a machine may too; a prohibition on inference from public information is an information ban with better branding. But the deeper problem is where the trend lines point: the dossier is ceasing to be a **file** and becoming a **query**. Today, compilation takes investigative effort, and that scarcity is what makes a dossier worth selling, worth seizing, worth regulating as a thing. As lives become digitally documented and agents transact for their principals, any entity — agency, employer, neighbor — will be able to generate a comprehensive profile on demand at no meaningful cost. Every object-layer rule then fails on its own terms: a sale restriction governs a market that scarcity no longer sustains; a retention limit deletes a document that can be regenerated the moment it is needed; a notice of existence informs people of what everyone should already assume; and a warrant to **acquire** asks the state to seek judicial permission for the same reading any citizen can do unaided — an imbalance, not a protection. The durable layer is the one that cannot be regenerated out of reach: the **outcome**. Deploying a dossier for harassment or stalking is already unlawful. Deciding against a person on the strength of an inferred profile without reasons and contest is what R11’s second tier reaches, and targeting a person for state action — the audit, the watchlist, the stop — is itself a use, carrying the same duties when it matures into an adverse decision, with inferred sensitive traits protected at use however the inference was made (R11). The state needs no permission to read what its citizens can read; it owes process when it acts. This is a deliberate and bounded position: the paper declines to regulate the compilation harms that are not outcomes — the chill of knowing a profile is possible — because that chill attaches to a capability that no possession rule can repeal, and says so rather than gesturing at coverage it does not provide. In the interim, while

compilation still takes effort and a dossier market still exists, other bodies of law police that market's abuses; this paper regulates the layer that will still be standing when the market is gone.

### New AI-specific collective-bargaining rights

Some propose statutory bargaining rights over workplace AI deployment. The analysis does not support a new, AI-specific channel, and the reason is the portfolio's own durability test: a bargaining right keyed to one technology is obsolete the day the technology is renamed, and the existing machinery [46] is at least tool-neutral. But the scope of that machinery should be stated accurately, not generously: it reaches **organized** workers — a minority of the workforce, with the gig and contractor workers most exposed to algorithmic management largely outside the statute — and doctrine reaches the **effects** of an automation decision more readily than the decision itself. The honest formulation is that existing law reaches organized workers' effects bargaining, and the reader can judge whether that is enough. What this portfolio adds, it adds without creating a bargaining channel: disclosure to workers of the algorithmic-management systems in use and limits on monitoring beyond the task (R3), contest rights over automated discipline and deactivation (R11 tier two), seats in R9's stress-test design and on R10's panel and R4's trigger steward — nominated by worker organizations, with votes, paid time, and published dissents — whistleblower protection reaching contractors in R7, co-financed and automatically triggered adjustment in R4, and the recognized fallback occupation of R5. Whether bargaining law itself should be amended is a question for labor law on its own terms, and this paper leaves it there.

### Settling AI's moral status by statute now

Some propose granting AI systems legal personhood; others propose legislating that AI can never warrant moral consideration. Both are premature. The first commits society on thin evidence. The second forecloses a question that evidence may later reopen,

and history shows the cost of assuming away the capacity to suffer [38]. The analysis supports pre-set evidence criteria and a standing expert panel (R10), which let policy respond to evidence as it accumulates. The stance has a formal pedigree: the literature on decision-making under moral uncertainty argues that an agent unsure among moral theories should keep options open where the stakes are asymmetric [69] — agnosticism here is not indecision but the rational wager.

### Retraining as the primary answer to job loss

Retraining is the traditional response to technological displacement. The analysis, however, finds that the technology devalues training itself, because it lets an untrained person do work that once took years to learn. The empirical record points the same way — the federal evaluation of the trade-adjustment program found its training-centered participants still earning less than comparable workers years into the follow-up [55]. Retraining remains useful, but the direct-support menu of R4 — wage insurance with employer co-financing, portable benefits, income support, early-retirement bridges, and public-service employment — is the more robust response.

## 11 Anticipating Objections

The recommendations will draw opposition from several directions. The concerns are legitimate, and we address the most significant in turn.

### "Regulation will stifle innovation and cede leadership to rivals."

The core portfolio is lighter than its length suggests, and Table 2 sets out what each measure costs and who pays: most measures are institutional investments (reserves, reporting systems, research) rather than restrictions on development, and the ones that do bind — R8's staging, R6's levy — carry small-entity tracks, exemption floors, and damage caps precisely so their fixed costs do not fall regressively on challengers. Competition oversight (R1) protects new entrants against

dominant firms, which is pro-innovation. No-fault compensation (R6) gives developers **predictable** liability exposure. The U.S. Supreme Court has described the vaccine injury program as created “to stabilize a vaccine market adversely affected by an increase in vaccine-related tort litigation” [56]. Aviation, which accepts extensive incident reporting and safety oversight, is among the most trusted industries in the world. The recommendations also reject the general slowdown that innovation advocates most fear (Section 10).

**“This does not go far enough. We should pause.”**

The analysis takes catastrophic risk seriously. Pathway A ends in correlated failure and lock-in, and Pathway B in a contest over control. But it finds that a general pause does not address the mechanism that makes those outcomes dangerous, which is loss of visibility, and that it may shift development toward less careful actors. The portfolio instead builds the capacity to act decisively: incident data to recognize danger (R7), circuit breakers to contain it (R9), evidence-triggered pauses to stop specific activities, and human fallback capacity to operate without AI if necessary (R5). For those focused on safety, these are the tools that still work when the stakes are highest.

**“This creates costly bureaucracy for speculative risks.”**

Several measures address problems that exist today: market concentration, resource demands, worker displacement, and misuse. Others, such as reserves, drills, and pre-set criteria, are inexpensive insurance, and the Collingridge dilemma implies that they are far cheaper now than later [29]. Pathway-specific measures are adopted only when signposts appear [48]. A good deal of the portfolio can build on existing institutions: utility planning and tariffs, reporting models, and financial-stability bodies.

**“No-fault compensation lets companies off the hook.”**

The fund is financed by an industry levy, so the industry pays. It is paired with strict operator liability, and fault-based claims remain available where a cause can be established. The purpose is to ensure that victims are compensated even when a cause cannot be proven, which is the situation in which fault-based systems leave them with nothing.

**“Identity requirements and monitoring threaten privacy and free expression.”**

The analysis deliberately favors controls on **physical inputs** over controls on information, which protects expression, and it treats anonymity as a value under two-sided attack. The technology makes a human’s anonymity easier to destroy — anything that has read the whole Internet can profile almost anyone — and easier to counterfeit, when one operator presents as ten thousand people. The portfolio’s principle: **anonymity is a protection for humans, not a property of scale**. A person may speak, buy, and associate unidentified; an operator running an automated crowd is not exercising anonymity but manufacturing false provenance, and owes the public one fact — that the crowd is automated (R13). The principle draws an asymmetry the careful reader will test, so it is argued rather than assumed: scale transforms the moral character of **speaking**, because ten thousand personas run by one operator deceive their audience about how many people are present — the harm is the deception, and it exists only at scale. Scale does not transform the moral character of **reading**, because no one is deceived by being read; what scale changes there is the consequences of **acting** on what was read, which is exactly where the portfolio’s use-based duties attach (R11, Section 10). On the human side, identity checks are confined to a closed, enumerated list of high-stakes transaction classes on the pseudoephedrine model [19] — with distributed records and a statutory ban on the centralized query systems that model historically grew [61], retention limits, legal-process access, an

expressive carve-out, suppression-plus-damages remedies, and a mandatory evidentiary sunset (Section 8) — and the answer to inferred dossiers is use-based: reasons and contest rights when a profile is used against a person (R11), with synthesis of public information left free.

### **“Workers will bear the costs of the transition.”**

Displacement falls on identifiable people, and a portfolio built around markets, inputs, and systemic risk could easily leave them to absorb it. The recommendations answer this with specific institutions rather than sympathy: displacement support on labor-data triggers under a worker-seated steward, co-financed by deploying employers on an observable base (R4); whistleblower protection, reaching contractors, for the near-miss reports the system runs on (R7); worker seats — nominated by worker organizations, voting, with paid time and published dissents — where stress-test scenarios are designed, evidence criteria are set, and displacement triggers are kept (R9, R10, R4); continuity rights, monitoring limits, and disclosure of the management systems in use when workplace AI changes under a worker’s feet (R3); contest rights over automated discipline and deactivation (R11); and a recognized, paid, organizable occupation in keeping manual capability alive (R5). The paper declines to create a job-protected right to refuse AI tools or a new bargaining channel for workers; Sections 7 and 10 give the reasons, and the bargaining section states the limits of existing law plainly rather than leaning on them.

### **“Pathway B is science fiction, and AI welfare is a distraction.”**

Pathway B is not a forecast. It is a path that could be built deliberately (Section 7), and the Pathway B measures recommended now are modest: public research and criteria agreed in advance. Pre-set criteria also guard against the opposite error of being manipulated by systems or companies into overstating AI’s inner life. The costs of error are asymmetric: history shows

that assuming no capacity for suffering has sometimes caused great harm [38].

### **“Controls will hurt national security.”**

Several recommendations strengthen security directly. Control of physical inputs is already national policy [51]. Manual fallback capacity protects critical infrastructure against both failures and attacks [32]. Systemic oversight reduces the chance that a single failure cascades across sectors. A nation whose critical systems can operate without AI is harder to coerce, not easier.

## **12 Conclusion**

Policymakers cannot know whether AI will become a tool that gradually acts on its own, a population that changes itself and may come to matter morally, both, or neither. This paper’s central claim is that they do not need to know in order to act well.

By examining two futures with AI deliberately removed, the analysis finds a consistent pattern. Both futures cross a threshold at which the causes of events can no longer be observed. Rules that depend on identifying who acted or why fail at that threshold. Rules that attach to physical inputs, named responsible parties, measurable effects, and the system as a whole continue to work. Those rules are largely unglamorous: competition, reserves, compensation, incident reporting, fallback capacity, research — and one reservation of principle, the decisions a perfect machine still should not make.

They also share a deadline. Each is inexpensive to adopt now and difficult to adopt later. The Collingridge dilemma is not a reason for fatalism. It is a reason to act before the need is obvious.

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**Choose policies that work on both pathways. Adopt them early. Do not rely on labeling the work or finding someone to blame.**

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## Appendix A: Implementation Notes

This appendix holds the implementation specifics the recommendations point to, so that the body text can carry principles and the mechanics can be revised without touching the argument.

**R2 — instrument-to-forum map.** The demand assessment runs as a joint industry–government process feeding the grid’s existing long-term reliability assessments and regional transmission planning; it directs no state plan. Firm-service and interruptible tariffs, and retail cost allocation, belong to state utility commissions; wholesale capacity obligations and interconnection terms to the system operators and the federal energy regulator; water to state water law and utility capacity processes; chips to the export-control and industrial-policy bodies that already govern them [51]. Qualified-load criteria: escalating study deposits, a contracted ramp schedule, disclosure of duplicate interconnection requests. Cost-causation instruments: long-term contracts with minimum-demand charges sized to contracted capacity, exit fees and collateral, direct assignment of interconnection-specific facilities. Capacity-contribution instruments (satisfiable off-site): contracted new-build generation, storage, or transmission co-investment, benchmarked in megawatts against firm draw. Tranche design: a single facility attests which workloads sit on firm and which on interruptible service, verified against R9’s dependency map; the yielding resource must be lawful to run at the called hour.

**R5 — funding channels.** Regulated utilities: rate-base treatment with pre-approved recovery. Merchant and competitive operators: capacity-market adders, tax credits, or cost-sharing grants. Drills scheduled with the reliability coordinator; simulator substitution where live degradation is unsafe.

**R8 — parameter defaults.** Capability threshold indexed to frontier training scale, updated by rule on a mandated cycle; expansion-review shot-clock of 90 days (120 for first-in-class deployments) with approval deemed granted on expiry; small entity defined by existing small-business practice, by employee count and revenue, not by model size.

## Appendix B: The Arithmetic of Precaution

Two arguments in the body rest on a simple expected-cost comparison. This appendix states it once, so that the body text can refer to it without carrying the variables.

**The welfare baseline (R16, Section 5).** Let  $p$  be a decision-maker’s credence that the beings in question can suffer. Let  $C$  be the cost of the welfare baseline. Let  $H$  be the harm if beings that can suffer are produced industrially without protection. The baseline is justified whenever

$$C < p \cdot H.$$

The inequality does not require believing that the beings can suffer. It requires only that the credence is not zero and that the harm, if real, is large. With a credence as low as one in twenty, for example, the baseline is justified whenever the unprotected harm would exceed twenty times the baseline’s cost. That is why the arithmetic favors even the skeptic.

**The second column.** The ledger has an opposite error, and the triggers price it. Let  $C'$  be the cost of crediting a performance: resources diverted on the strength of distress that was manufactured rather than felt. Under R16, protection scales with evidence measured against criteria fixed in advance, not with sentiment. That caps  $C'$ , because no amount of performed distress moves a protection that the pre-set evidence does not support.

**What changes over time.** The inequality does not weaken as the technology matures. If anything,  $H$  grows with the population at stake. What grows with installed capital is the **resistance** to acting on it. The comparison still favors the baseline in the late stages. Enacting the baseline is what stops being possible. This is the Collingridge dilemma in numerical form.

**Duplication (Section 7).** A pill augments one person, and a chamber is one chamber. An AI model can be copied without limit at near-zero cost. Duplication enters both sides of the comparison. It multiplies the population over which  $H$  runs, since a welfare harm in one model is a harm in every copy. It also multiplies the deployments over which R6's levy and R8's monitoring must scale, which raises the costs on the other side. Duplication therefore makes getting the comparison right more important, whichever way it comes out.

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